GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT PATHOLOGIES VIA PROJECTION NORMALIZATION, AND THE INTRODUCTION OF IFOPNG

JAKUB MICHAŁOWSKI

THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE

DEPARTMENT OF
COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE
SCHOOL OF HUMANITIES AND DIGITAL SCIENCES
TILBURG UNIVERSITY

STUDENT NUMBER

u253954

COMMITTEE

dr. Giacomo Spiegler
dr. The Second Reader

LOCATION

Tilburg University
School of Humanities and Digital Sciences
Research Center of Cognitive Science &
Artificial Intelligence
Tilburg, The Netherlands

DATE

June 3, 2026

WORD COUNT

≈7800

ACKNOWLEDGMENTS

The author thanks dr. Giacomo Spiegler for supervision and feedback throughout this project.

GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT PATHOLOGIES VIA PROJECTION NORMALIZATION, AND THE INTRODUCTION OF IFOPNG

JAKUB MICHAŁOWSKI

Abstract

This thesis addresses the fundamental incompatibility of applying gradient projection-based continual learning (CL) methods to hypernetwork architectures, establishing a novel framework for stable subspace optimization. Traditional projection methods collapse due to a structural–numerical pathology where the architectural necessity of weight-chunking, required to avoid an output parameter explosion, forces the model to be produced in hundreds or thousands of sequential chunks before a single projection step can execute. This repetitive chunk-generation loop accumulates an exceptionally large gradient volume within the memory matrix, blowing up the condition number of the projection correlation matrix to infinity and causing catastrophic inversion failures that the standard regularization parameter λ cannot rescue.

The primary research objective is to eliminate these numerical singularities and systematically evaluate whether combining hard geometric projections on top of task-conditioned hypernetworks and their output-scoped functional regularizers yields a constructive synergy. To achieve this, we introduce *projection-scoped normalization*: gradient columns stored in the memory matrix G are rescaled to unit ℓ_2 norm, and the diagonal empirical Fisher vector is rescaled by its absolute maximum, together stabilizing the metric tensor within the local projection algebra. We also introduce `iFOPNG`, an inertial variant of Fisher-Orthogonal Projected Natural Gradient Descent (Garg, Kolhe, Peng, & Gopalam, 2026) that replaces the current-task Fisher F_{new} with the combined curvature $F_c = F_{\text{new}} + F_{\text{old}}$, inspired by the parameter-inertia concept of Elastic Weight Consolidation. Methods are benchmarked on sequential Split and Permuted MNIST, Split-CIFAR10 and Split-CIFAR100. The empirical findings reveal that projection normalization completely resolves matrix inversion collapse, and that in the hypernetwork setting all normalized projection methods outperform plain SGD, with `iFOPNG` matching the best projection baseline. However, a standalone hypernetwork optimized with `Adam` and von Oswald's functional regularizer consistently surpasses all projection variants, indicating that momentum, which projection methods forgo by design, is the decisive factor in compressed generative weight spaces.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source. The MNIST (owned by LeCun et al.) and CIFAR (CIFAR-10 and CIFAR-100, owned by Krizhevsky et al.) datasets are public, anonymized datasets licensed for research. These datasets form the basis for the Split-MNIST, Permuted MNIST, Split-CIFAR10, and Split-CIFAR100 benchmarks evaluated in this work. No human or animal data was collected. Data was acquired via the `torchvision.datasets` API.

FIGURES. All figures and plots were created entirely by the author using `matplotlib` and Weights & Biases.

CODE. The baseline FOPNG framework was adapted from Garg et al. (2026); all other execution infrastructure was implemented by the author (`https://github.com/kelares/HyperFisher`). Libraries used: Python (v3.10), PyTorch (v2.x), Torchvision, Weights & Biases, NumPy, Matplotlib, and Tqdm.

TECHNOLOGY. The manuscript was typeset in LaTeX using BibTeX. During the preparation of this work, the author(s) used Gemini and Claude in order to optimize academic phrasing, outline structure, paraphrase transitions, and check grammar. After using these tools/services, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the thesis. AI Citations: Google. (2026). *Gemini* [Large language model]. https://gemini.google.com; Anthropic. (2026). *Claude* [Large language model]. https://claude.ai.

2 INTRODUCTION

The capacity for continuous, sequential adaptation without the eradication of historical knowledge remains an architectural cornerstone of biological intelligence, yet it presents a fundamental barrier for artificial neural networks. In deep learning paradigms, sequential training across non-stationary data distributions triggers a systemic pathology known as *catastrophic forgetting* (McCloskey & Cohen, 1989). When a network optimizes its parameters for an incoming task, gradient updates blindly overwrite the weight configurations essential for retaining prior task spaces, shifting the internal representation manifolds away from historical objective minima.

Modern continual learning (CL) literature is conventionally structured into three core algorithmic families: replay-based strategies that leverage episodic data buffers, regularization-based frameworks that penalize parameter deviations, and architectural or parameter-isolation methods (De Lange et al., 2022). Within the meta-learning domain, hypernetworks (Ha, Dai, & Le, 2016) have emerged as an elegant paradigm. Hypernetworks circumvent parameter-space conflicts by utilizing a centralized, lightweight meta-model to generate the target network's complete parameter weights conditioned on task embeddings, as formalized by Oswald, Henning, Grewe, and Sacramento (2020). Concurrently, within the field of geometric optimization, gradient projection methods—exemplified by the recently proposed Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026)—explicitly constrain subsequent updates to the null-space of historical parameter spaces.

Despite the mathematical elegance of both hypernetworks and gradient projection methods, their structural intersection uncovers a critical and previously unexamined optimization pathology. To prevent a catastrophic parameter explosion within the meta-model, a hypernetwork must use sequential weight-chunking. Depending on the dimensional footprint of the target architecture, this constraint forces the target model to be produced in hundreds or thousands of individual sub-blocks. Executing this spawning mechanism prior to a single backward pass accumulates gradients additively, resulting in parameter gradients of exceptionally high magnitude. This explosion numerically erases the stabilizing effect of λ during matrix inversion, inflating the condition number of the projection kernel to extreme scales (e.g., exceeding 10^{10}) and inducing catastrophic projection failures.

Research Objectives and Algorithmic Framework

The primary research objective of this thesis is to systematically resolve this linear-algebraic collapse, map the initialization mechanics of meta-projection pipelines, and evaluate whether the integration of hard geometric constraints on top of soft functional regularizers yields an optimized sequential learning system.

To achieve this, we introduce projection-scoped normalization (columnwise ℓ_2 orthonormalization of G combined with absolute-maximum scaling of $\hat{F}$), an `AdamW` first-task initialization protocol, and `iFOPNG`—an inertial FOPNG variant using the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$. These contributions enable stable gradient projection in chunked hypernetworks, with direct applicability to on-device sequential learning where retaining adaptive momentum is critical.

Research Questions

To systematically navigate these boundaries, this thesis addresses the following primary research question:

> *To what extent can gradient projection methods be successfully integrated into chunked hypernetwork architectures for continual learning, and how do their structural pathologies, initialization protocols, and inertial variants impact optimization efficiency?*

This is decomposed into four sub-research questions (Sub-RQs):

Sub-RQ1 **(Synergy vs. Conflict):** Does the joint integration of gradient projection frameworks and soft output-scoped functional regularization yield a constructive synergy, or does a standalone regularized hypernetwork offer a more efficient optimization frontier?

Sub-RQ2 **(Gradient Pathology):** How does the architectural necessity of spawning thousands of sequential weight-chunks impact gradient magnitudes, and to what extent can local projection-scoped normalization prevent the resulting explosion from collapsing the matrix inversion?

Sub-RQ3 **(Parameter Inertia):** To what degree does evaluating the natural gradient projection under a joint Riemannian metric ($F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$), rather than the task-specific metric ($\hat{F}_{\text{new}}$) used in standard FOPNG, influence task retention?

Sub-RQ4 **(Fisher Accumulation):** How should the historical component $\hat{F}_{\text{old}}$ be accumulated across tasks? Specifically, does a non-decaying maximum operator preserve early-task knowledge more effectively than an exponential moving average (EMA)?

Summary of Empirical Findings

Projection-scoped normalization completely resolves the matrix inversion collapse; `iFOPNG` consistently outperforms `FOPNG` wherever inter-task Fisher overlap is substantial; and the MAX accumulation operator is the preferred hyperparameter-free default. Critically, normalized projection methods nonetheless fail to match a hypernetwork optimized via `Adam`: the loss of adaptive momentum proves to be the binding constraint in compressed generative weight spaces.

3 RELATED WORK

This section builds the theoretical and empirical context for the thesis, tracing a path from foundational continual learning challenges through regularization and projection methods, to hypernetwork architectures and the structural gap at their intersection.

3.1 *Continual Learning: Taxonomy and the Stability-Plasticity Dilemma*

Modern machine learning systems are predominantly trained on static, i.i.d. datasets — a condition that rarely holds in deployment, where agents must adapt to non-stationary data streams while retaining prior knowledge. The fundamental tension between these two objectives is formalized as the *stability-plasticity dilemma* (Grossberg, 1980): an agent must remain plastic enough to absorb novel distributions yet stable enough to resist the erasure of established representations. When standard architectures trained by empirical risk minimization are exposed to sequential data, they suffer *catastrophic forgetting* (McCloskey & Cohen, 1989) — abrupt and near-total loss of previously learned capabilities upon exposure to new task data.

The continual learning literature addresses this across three canonical scenarios (De Lange et al., 2022): *task-incremental learning*, where the task identity is available at both training and test time; *class-incremental learning*, where task identity is withheld at inference; and *domain-incremental learning*, where the output space is shared but the input distribution shifts continuously. This thesis focuses on the task-incremental setting, the scenario in which projection-based methods and hypernetwork architectures have been most thoroughly evaluated and in which structural comparisons between them are most tractable.

3.2 *Evaluation Protocols: Multi-Head and Single-Head Designs*

Within the task-incremental setting, two evaluation protocols govern how the network's output layer is organized. In the *multi-head* protocol, each task is assigned a dedicated output head; at test time the correct head is selected using the known task identity. This eliminates output-level interference between tasks but requires that task identity be available at inference time, a condition that may be unrealistic in deployment. In the *single-head* protocol, all tasks share one output layer, requiring the backbone alone to resolve inter-task interference — a substantially harder problem.

Farquhar and Gal (2018) demonstrated that multi-head evaluations systematically inflate apparent method performance by removing a major source of forgetting, and argued that single-head evaluation is the more rigorous protocol for assessing continual learning efficacy. This distinction has direct implications for how gradient projection methods and hypernetworks are evaluated: projection methods were predominantly developed and benchmarked in multi-head settings, while hypernetworks implement an implicit form of task separation through their generative structure.

3.3 *Regularization-Based Approaches*

The dominant paradigm for addressing catastrophic forgetting is *parameter regularization*, of which Elastic Weight Consolidation (EWC) (Kirkpatrick et al., 2017) is the canonical representative. EWC estimates the structural importance of each weight using the diagonal of the empirical Fisher Information Matrix $\hat{F}$, and applies a soft quadratic penalty that resists changes to parameters important to prior tasks:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i \left(\theta_i - \theta^*_{i,t-1}\right)^2. \tag{1}$$

EWC is computationally efficient and broadly applicable, and serves as the standard reference point against which new continual learning methods are measured. Its principal limitation is that soft penalties permit bounded parameter drift: cumulative updates across a long task sequence gradually erode the protection of earlier task spaces, a failure mode that becomes increasingly pronounced as T grows (Farquhar & Gal, 2018).

A related limitation concerns baseline comparisons. The superiority of complex regularization schemes over vanilla optimizers is frequently overstated when those baselines are under-tuned. When SGD and Adam baselines are tuned to their optimal hyperparameter regime, they often present a competitive threshold that challenges or matches more sophisticated approaches — an empirical reality that our baseline evaluations directly address.

3.4 *Gradient Projection Methods*

To enforce strict, non-negotiable protection for prior knowledge rather than relying on penalties that can be overcome by sufficiently large gradients, a second paradigm uses explicit *gradient projection*. Orthogonal Gradient Descent (OGD) (Farajtabar, Azizan, Mott, & Li, 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K]$ of historical task gradient directions and projects incoming gradients onto the orthogonal complement of $\mathrm{Col}(G)$ before each update:

$$\tilde{g} = g - G(G^\top G)^{-1} G^\top g. \tag{2}$$

This provides a hard guarantee: the update cannot increase the loss on previously seen tasks. Generalization bounds established within the Neural Tangent Kernel regime confirm that OGD provides exact protection in linear and mildly nonlinear settings (Bennani, Doan, & Sugiyama, 2020).

The mathematical appeal of projection methods over EWC lies precisely in this hardness: where EWC applies a soft penalty that can be overwhelmed, projection methods impose a geometric constraint that is either satisfied or not. However, standard Euclidean projection treats the parameter space as flat and assigns equal importance to all directions. Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026) addresses this by performing the projection in the Riemannian geometry defined by the Fisher metric, so that the orthogonality constraint respects the curvature of the loss surface. The related Fisher Natural Gradient (FNG) (Garg et al., 2026) applies the natural gradient without projection, and Orthogonal Natural Gradient (ONG) (**?**) combines natural gradient preconditioning with Euclidean projection — sitting structurally between OGD and FOPNG.

3.5 *Hypernetworks for Continual Learning*

An orthogonal architectural response to catastrophic forgetting replaces the multi-head output design entirely with a *hypernetwork* (Ha et al., 2016): a meta-model $\mathcal{H}(\cdot;\Theta_h)$ trained to synthesize the complete weight vector $\theta_t = \mathcal{H}(e_t;\Theta_h)$ of a target network, conditioned on a task embedding e_t. Each task receives a unique weight configuration generated on demand, rather than a dedicated output head, achieving implicit task separation through the generative bottleneck and impressive parameter compression ratios relative to storing separate networks per task. Generating the full θ_t in a single forward pass is computationally intractable for large target networks, so Oswald et al. (2020) introduced *weight chunking*: the generator iterates over a secondary chunk embedding c_k, producing C_s target parameters per pass and assembling the full weight vector by concatenation. To prevent the hypernetwork from forgetting how to generate prior task configurations, a soft *functional regularizer* is applied in the output space of the generator:

$$\mathcal{L}_{\text{reg}} = \beta \sum_{s=1}^{t-1} \left\| \mathcal{H}(\Theta_h, e_s) - \theta_s^* \right\|_2^2, \tag{3}$$

where θ_s^* denotes the weight snapshot saved after task s. This penalty can be interpreted as EWC applied at the meta level: rather than anchoring individual network parameters, it anchors the generated weight configurations for each previously seen task, penalizing drift in the space where task-relevant quantities actually live. Recent extensions have refined the hypernetwork approach: Partial Hypernetworks (Hemati, Lomonaco, Bacciu, & Borth, 2023) reduce the computational overhead of full weight generation, and Prototype Augmented Hypernetworks (Fuente et al., 2025) augment task embeddings with prototype representations to further stabilize sequential learning.

3.6 *Identified Gaps and Motivating Pathologies*

Oswald's finding raises a critical and unanswered question: if EWC-style *soft parameter penalties* are insufficient for hypernetwork continual learning, do the *hard geometric constraints* of gradient projection methods — which operate through a fundamentally different mechanism — fare differently? Projection methods do not penalize the hypernetwork's parameter values; they constrain the *direction* of the generator's updates. Whether this directional constraint is compatible with the hypernetwork's generative structure, or introduces its own failure modes, has not been investigated. This intersection is the primary research contribution of this thesis. We identify four specific structural pathologies that arise at this intersection and that the experiments are designed to address.

1. FUNCTIONAL-PARAMETRIC INTERFACE CONFLICT (SUB-RQ1). Layering hard parameter-space projections onto a hypernetwork that already employs soft output-space regularization creates a potential structural conflict: the projection constraint strips momentum from the generator's updates, while the functional regularizer simultaneously pulls them toward prior weight configurations. Whether this joint constraint regime yields a constructive synergy or over-constrains the generator's plasticity remains an open empirical question. A secondary structural dimension of this conflict concerns the seeding of the projection subspace. The gradient memory G and Fisher estimate $\hat{F}$ are collected entirely from first-task training; standard `Adam` folds L_2 weight decay into the gradient estimate, contaminating both G and $\hat{F}$ with a weight-magnitude component unrelated to task-loss curvature (Hu, Yu, Cheng, Liu, & Song, 2026). `AdamW`'s decoupled weight decay preserves gradient geometric purity, and whether this yields a measurably cleaner initial projection basis is examined as a secondary diagnostic within Sub-RQ1.

2. CHUNK-INDUCED GRADIENT ACCUMULATION PATHOLOGY (SUB--RQ2). The architectural necessity of spawning hundreds or thousands of sequential weight-chunks before a single projection step executes causes gradient vectors and Fisher estimates to accumulate to extreme magnitudes, inflating the condition number of the projection kernel $G^\top \hat{F} G$ and causing

matrix inversion failures that standard regularization parameters cannot mitigate. This pathology is unique to the hypernetwork setting and has no analogue in standalone network experiments.

3. PROJECTION RIGIDITY AND ELASTIC PARAMETER INERTIA (SUB--RQ3). Rigid projection baselines enforce an absolute boundary that completely locks historical gradient directions, stripping the optimizer of degrees of freedom that may be needed for efficient new-task learning. We address this rigidity through `iFOPNG`, an elastic variant of FOPNG introduced in this thesis, which replaces the hard projection boundary with a combined Fisher preconditioner $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$. Rather than imposing a hard constraint, F_c damps movement in directions important to *any* prior task via the metric geometry, providing an EWC-style elastic inertia boundary without a penalty term.

4. FISHER ACCUMULATION OPERATOR AND LONG-HORIZON PLASTICITY (SUB-RQ4). The historical Fisher $\hat{F}_{\text{old}}$ underlying the `iFOPNG` elastic boundary is maintained via a non-decaying maximum operator: $\hat{F}_{\text{old}} \leftarrow \max(\hat{F}_{\text{old}}, \hat{F}_{\text{new}})$. This operator guarantees that any parameter ever deemed important retains that importance permanently, providing strong protection for early-task knowledge. Whether this monotonic accumulation imposes a measurable plasticity cost over long task sequences — relative to a decaying exponential moving average — and whether the two operators trade off forgetting against new-task learning differently across sequence lengths, is the subject of Sub-RQ4.

4 METHODOLOGY

This section specifies the precise formulation of every method and architectural component used in our experiments, providing sufficient detail for reproducibility. Conceptual motivation and positioning relative to prior work are covered in Section 3; we restrict our attention here to the exact algorithmic instantiations evaluated.

4.1 *Continual Learning Protocol*

4.1.1 *Task Setup and Evaluation Metrics*

A continual learning agent receives a sequence of T tasks $\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_T$ presented in order. At task t, only the data from $\mathcal{T}_t$ is available; prior task data is not retained. After each task is trained, the model is evaluated on all tasks seen so far. We report two aggregate metrics: *average accuracy*

$$A_t = \frac{1}{t}\sum_{i=1}^{t} a_{t,i}, \tag{4}$$

and *backward transfer*

$$\mathrm{BWT} = \frac{1}{T-1}\sum_{t=2}^{T}(a_{T,t} - a_{t,t}), \tag{5}$$

where $a_{i,j}$ denotes accuracy on task j immediately after training task i. Negative BWT quantifies catastrophic forgetting; a value of zero indicates perfect retention.

4.1.2 *Multi-Head and Single-Head Evaluation*

Following Farquhar and Gal (2018), who demonstrated that multi-head evaluations can inflate apparent method performance, we report results under both protocols for MNIST-based benchmarks.

4.2 *Baseline and Regularization Methods*

4.2.1 *Fisher Information Matrix*

All Fisher-based methods in our pipeline use the *diagonal empirical approximation*

$$\hat{F}_i \approx \frac{1}{N}\sum_{n=1}^{N}\left(\frac{\partial \mathcal{L}^{(n)}}{\partial \theta_i}\right)^2, \tag{6}$$

estimated over N held-out samples at the end of each task. The full FIM is intractable for our network sizes; the diagonal approximation is standard across all evaluated methods (Garg et al., 2026; Kirkpatrick et al., 2017).

4.2.2 *Natural Gradient Descent*

Natural gradient descent (Amari, n.d.) preconditions the gradient with the inverse FIM to respect the curvature of the statistical manifold:

$$\Delta\theta_{\mathrm{nat}} = -\eta\, \hat{F}^{-1}\nabla_\theta \mathcal{L}. \tag{7}$$

With a diagonal Fisher approximation, inversion reduces to element-wise scaling and remains computationally tractable. The natural gradient underpins FOPNG, FNG, and ONG, which extend it with projection or trust-region constraints.

4.2.3 *Elastic Weight Consolidation*

EWC (Kirkpatrick et al., 2017) augments the task loss with a Fisher-weighted quadratic penalty:

$$\mathcal{L}_{\mathrm{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2}\sum_i \hat{F}_i\,(\theta_i - \theta^*_{i,t-1})^2, \tag{8}$$

where θ^*_{t-1} denotes the parameter snapshot after task $t-1$. We use $\lambda = 10^{-3}$ for all hypernetwork configurations and reproduce the per-benchmark values from Garg et al. (2026) Table 1 for standalone evaluations. Fisher estimates are computed over the full training set for MNIST-based benchmarks and over $N=1024$ samples for CIFAR benchmarks.

4.3 *Gradient Projection Methods*

4.3.1 *Orthogonal Gradient Descent*

OGD (Farajtabar et al., 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K] \in \mathbb{R}^{p\times K}$ of past task gradient directions and projects the current gradient onto the orthogonal complement of $\mathrm{Col}(G)$:

$$\tilde{g} = g - G(G^\top G)^{-1}G^\top g. \tag{9}$$

OGD operates in Euclidean space, treating all parameter directions symmetrically.

GRADIENT MEMORY MANAGEMENT. In all projection methods, G is maintained as a fixed-budget basis of at most M columns. Garg et al. (2026) set $M = T \times K$ across all benchmarks, so compression never triggers in their experiments. We address this unexamined regime explicitly: incoming gradient batches are appended directly to G; when the column count exceeds M, the full basis is recompressed via truncated SVD and $G \leftarrow U_{:,1:M}$, retaining the M directions of greatest variance in the accumulated gradient memory. When projection-scoped normalization is enabled, QR decomposition is additionally applied to each incoming batch at insertion time (Section 4.5.1).

4.3.2 *Orthogonal Natural Gradient and Fisher Natural Gradient*

ONG applies OGD-style projection to the natural gradient rather than to the raw gradient, combining Fisher preconditioning with a Euclidean orthogonality constraint:

$$\Delta\theta_{\mathrm{ONG}} = -\eta\left(\hat{F}^{-1}g - G(G^\top G)^{-1}G^\top\hat{F}^{-1}g\right). \tag{10}$$

Unlike FOPNG, the projection is measured in Euclidean rather than Fisher metric space, so the orthogonality guarantee does not extend to the statistical manifold.

FNG (Garg et al., 2026) applies no projection whatsoever; it is the trust-region normalized natural gradient:

$$v_{\text{FNG}} = \eta \, \frac{\hat{F}^{-1} g}{\sqrt{g^\top \hat{F}^{-1} g}}. \tag{11}$$

FNG serves as a curvature-aware SGD baseline on the Riemannian manifold, isolating the contribution of Fisher preconditioning without any subspace constraint. Both methods use the same memory budget as OGD and FOPNG to ensure fair comparison.

4.3.3 *Fisher-Orthogonal Projected Natural Gradient Descent*

FOPNG (Garg et al., 2026) executes projection in the Riemannian metric defined by the Fisher:

$$\Delta\theta = -\eta \left(I - G(G^\top \hat{F} G)^{-1} G^\top \hat{F} \right) \hat{F}^{-1} \nabla_\theta \mathcal{L}_t, \tag{12}$$

where the regularization parameter λ is added to the diagonal of $G^\top \hat{F} G$ before inversion for numerical stability. We reproduce the per-benchmark learning rates and λ values from Garg et al. (2026) Table 1 for standalone configurations, and use a uniform lr=10^{-3}, λ=10^{-3} for all hypernetwork configurations.

4.3.4 *Inertia FOPNG (iFOPNG)*

In FOPNG, the parameter update takes the form

$$v = c \, \hat{F}_{\text{new}}^{-1} (g - u), \tag{13}$$

where $\hat{F}_{\text{new}}$ serves as the Riemannian metric: both the trust-region constraint $\|v\|_{\hat{F}_{\text{new}}} = \eta$ and the projection objective $\|g - \hat{F}_{\text{new}}^{-1} u\|^2_{\hat{F}_{\text{new}}}$ are measured in the current task's Fisher geometry. This is natural when no prior tasks exist, but it creates an asymmetry once history accumulates: the constraint subspace $\text{Col}(G)$ is built from gradient directions collected under $\hat{F}_{\text{old}}$, yet the projection metric assigns zero curvature to parameters that were important in previous tasks but are uninformative on the current one.

`iFOPNG` is introduced in this thesis as a variant of FOPNG that resolves this asymmetry by adopting the *combined metric*

$$F_c \triangleq \hat{F}_{\text{new}} + \hat{F}_{\text{old}}, \tag{14}$$

where $\hat{F}_{\text{old}}$ is maintained as the element-wise maximum of all per-task Fisher diagonals accumulated so far. Replacing (13) with $v = c\,F_c^{-1}(g-u)$, the constrained trust-region update becomes

$$v^* = -\eta\,\frac{F_c^{-1}Pg}{\sqrt{(Pg)^\top F_c^{-1}Pg}}, \qquad P = I - \hat{F}_{\text{old}}\,G\,A^{-1}\,G^\top \hat{F}_{\text{old}}, \tag{15}$$

where $A = G^\top \hat{F}_{\text{old}}\,F_c^{-1}\,\hat{F}_{\text{old}}\,G$. A formal derivation is provided in Appendix A.

The diagonal matrix F_c defines a new Riemannian metric on parameter space in which the effective mass of coordinate i is $(F_c)_i = (\hat{F}_{\text{new}})_i + (\hat{F}_{\text{old}})_i$. Parameters that were important to any previously seen task carry elevated $\hat{F}_{\text{old},i}$, suppressing the natural-gradient step size in those directions. Crucially, this suppression is not imposed through a penalty term; it emerges from the geometry of the optimisation problem itself. We call this property *parameter inertia*: a large cumulative Fisher importance makes a coordinate heavier and therefore harder to displace, in exact analogy to Newton's second law.

Note that the constraint subspace retains $\hat{F}_{\text{old}}$ as its defining geometry, since $\mathrm{Col}(G)$ encodes where previous tasks concentrated their gradient mass. Using F_c for the ambient metric while keeping $\hat{F}_{\text{old}}$ in the constraint induces an *oblique* projection: the component u is removed from g in the $\hat{F}_{\text{old}}$-weighted direction of the historical subspace, but the residual Pg is preconditioned by F_c^{-1} and its magnitude is evaluated in the F_c norm. The constraint describes *where* previous tasks lived; the metric describes the *combined curvature* under which the update should be invariant.

This design stands in direct contrast to Elastic Weight Consolidation (Kirkpatrick et al., 2017), which appends an explicit quadratic term

$$\mathcal{L}_{\text{EWC}} \;=\; \mathcal{L}_{\text{new}}(\theta) + \frac{\lambda}{2}\sum_i (\hat{F}_{\text{old}})_i\left(\theta_i - \theta_i^*\right)^2$$

to the loss function, applying a spring force that pulls parameters toward a reference point θ^*. `iFOPNG` applies no such force and maintains no reference point: a parameter that mattered before is not pulled back, it is simply made harder to displace, in proportion to how jointly curved the loss landscape is under both the old and new task distributions.

4.4 *Hypernetwork Architecture*

4.4.1 *Chunked Hypernetworks*

A hypernetwork (Ha et al., 2016) is a meta-model $\mathcal{H}(\cdot;\Theta_h)$ that generates the complete weight vector $\theta_t = \mathcal{H}(e_t;\Theta_h)$ of a target network conditioned on a task embedding e_t. Generating θ_t in a single forward pass would require an output layer of dimension $|\theta_t|$, which is computationally prohibitive for any non-trivial target architecture. *Weight chunking* (Oswald et al., 2020) resolves this by iterating over a secondary chunk embedding c_k, $k = 1, \ldots, K$, and producing a block of C_s target parameters per pass; the full weight vector is assembled by concatenation. Chunking makes the hypernetwork parameter-efficient but introduces the chunk-induced gradient accumulation pathology addressed in Section 4.5.1.

4.4.2 *Functional Regularizer*

The functional regularizer of Oswald et al. (2020) is active for all hypernetwork configurations, with regularization strength β treated as a tunable hyperparameter (see Appendix 8).

4.4.3 *Target Network and Dropout*

Standalone configurations reproduce the target network architecture of Garg et al. (2026) without modification, including two `Dropout`(p=0.5) layers in the `SimpleCIFARCNN` classifier head. Dropout is removed from the target network in all hypernetwork configurations: `PyTorch`'s `functional_call` propagates the parent module's `.training` mode to the target submodule, causing stochastic masking across all chunk-generation passes during training. The generative bottleneck ($\Theta_h : d_t{+}d_c \to d_h \to C_s$) and the functional regularizer $\mathcal{L}_{\text{reg}}$ provide sufficient implicit regularization in its place.

4.5 *Novel Algorithmic Contributions*

4.5.1 *Projection-Scoped Normalization*

When a hypernetwork spawns K sequential weight-chunks before a single projection step executes, the accumulated gradient vector and the diagonal Fisher entries grow in magnitude proportionally to K, inflating the condition number of the projection kernel $G^\top \hat{F} G$ and causing the matrix inversion in Equation (12) to fail catastrophically for any practically sized target network.

We resolve this with two independent rescaling operations that address distinct sources of ill-conditioning:

1. **Gradient basis orthonormalization.** Each batch of K gradient vectors collected after task t is replaced by the orthonormal factor Q from its thin QR decomposition before insertion into G:

$$[g_{t,1} \mid \cdots \mid g_{t,K}] \xrightarrow{\text{QR}} Q, \qquad Q^\top Q = I. \tag{16}$$

This operation applies *at storage time* and affects only the basis G; the gradient g entering the projection update itself is the unmodified parameter gradient probed directly from the model.

2. **Metric-tensor normalization.** Within each projection computation, both Fisher terms are divided by the maximum entry of the metric tensor F_c that defines the constrained natural gradient problem:

$$\tilde{F} = \frac{F}{\max_i (F_c)_i + \varepsilon}, \qquad F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}, \tag{17}$$

where F_c is the combined Fisher used as the Riemannian metric in iFOPNG, or $\hat{F}_{\text{new}}$ alone for FOPNG. This scaling is applied locally within the kernel computation and does not affect the optimisation trajectory outside the projection step.

Together, these operations ensure $G^\top \tilde{F} G$ remains well-conditioned regardless of the number of chunks spawned per task. Gradient orthonormalization eliminates column-scale inflation in G; metric-tensor normalization bounds the Fisher entries to $[0, 1]$, preventing the kernel from becoming numerically singular. Subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation alters the geometry of the projection.

PRINCIPLED RESCALING AND OPEN PROBLEM. The proposed normalization — QR orthonormalization of the gradient basis combined with max-entry scaling of F_c — is heuristic in nature: it discards the eigenvalue structure of the pullback metric and applies isotropic rather than curvature-aware rescaling to the chunk embedding directions. The principled alternative is the natural gradient under the Riemannian metric induced by the chunk-generating map:

$$\Delta c_i = -\eta \, (J_i^\top J_i + \lambda I)^{-1} \, \nabla_{c_i} \mathcal{L}, \qquad J_i = \frac{\partial W_i}{\partial c_i} \in \mathbb{R}^{C_s \times d_c}, \tag{18}$$

which rescales each chunk embedding direction in proportion to how strongly it moves the generated weights, and is tractable because $d_c \ll C_s$. This preconditioner was implemented and evaluated experimentally. Applied to the raw gradient *prior* to the projection step, it consistently degraded continual learning performance — BWT worsened by approximately 0.33 on Split-MNIST SH relative to the heuristic baseline. We attribute this to a metric mismatch: the FOPNG projection operator is calibrated for gradients in Euclidean parameter space, and preconditioning only the chunk embedding component before projection produces a geometrically inconsistent input to the kernel $G^\top \hat{F} G$. The principled resolution — applying $(J_i^\top J_i + \lambda I)^{-1}$ to the projected update v^* rather than to the raw gradient, so that the projection sees Euclidean gradients and the Riemannian step follows — is left to future work.

4.6 *Experimental Design*

The four sub-research questions demand qualitatively different experimental designs. Sub-RQ1 and Sub-RQ3 require head-to-head benchmark comparisons — projection methods against an unconstrained baseline, and `iFOPNG` against `FOPNG` respectively. Sub-RQ2 requires a controlled ablation over normalization conditions. Sub-RQ4 requires a long-horizon comparison of two Fisher accumulation operators on the same elastic preconditioner. Hyperparameter choices fall into two categories: those inherited directly from the literature, which require no further justification, and those specific to the novel hypernetwork experiments, each of which is justified below.

4.6.1 *Hyperparameter Provenance*

STANDALONE TARGET-NETWORK EXPERIMENTS. All hyperparameter values for experiments without a hypernetwork — learning rates, regularization coefficients, gradient budgets, Fisher sample counts, batch sizes, and epoch counts — are taken without modification from Garg et al. (2026) Table 1. `iFOPNG` uses the same values as `FOPNG`, since the elastic combined Fisher introduces no hyperparameters beyond those FOPNG already defines. `ONG` (**?**) is treated as a composite method: it applies `OGD`'s Euclidean projection operator to the natural gradient direction and introduces no hyperparameters beyond those already specified for `OGD` and `FNG`; its values therefore follow `OGD`'s Table 1 entries. First-task training follows the paper's protocol: SGD at the method's own learning rate for MNIST-based benchmarks, and SGD at a fixed rate of 10^{-3} for CIFAR-based benchmarks. No tuning was performed on these values; they are reproduced to enable direct numerical comparison with the published results.

HYPERNETWORK EXPERIMENTS. The hypernetwork experiments introduce settings not covered by Garg et al. (2026), which did not evaluate projection methods in a hypernetwork setting. The functional regularizer and the task and chunk embedding dimensions follow Oswald et al. (2020), the canonical hypernetwork continual learning reference; benchmark-specific values are listed in Appendix 8. Fisher estimation is capped at 1,024 samples rather than using the full training set: a full estimate on a hypernetwork requires one complete K-chunk generation pass per training sample, which is computationally intractable within the available compute budget. The cap of 1,024 is consistent with the FOPNG paper's own CIFAR-10 setting and is a standard practical approximation (Kirkpatrick et al., 2017). The gradient budget is set to $K = 80$ gradients per task, matching the FOPNG paper's CIFAR values, as no principled reason exists to increase it for the hypernetwork's similarly-sized target parameter space. Unlike the standalone experiments, the hypernetwork runs do exceed the gradient memory budget; compression is handled by the QR-and-SVD memory management scheme described in Section 4.3. First-task training for all hypernetwork experiments uses `AdamW`; this follows the Sub-RQ1 secondary diagnostic rationale described in Section 3, ensuring that gradient memory G and Fisher estimate $\hat{F}$ are seeded from a geometrically clean first-task basis rather than one contaminated by coupled L_2 weight decay.

UNIFORM LEARNING RATE FOR HYPERNETWORK EXPERIMENTS. Rather than using the method-specific learning rates of Table 1, all methods in the hypernetwork experiments use a uniform learning rate of 10^{-3}. The goal of these experiments is not to replicate Table 1 numbers — which apply to standalone networks — but to isolate the projection constraint as the sole independent variable. Method-specific rates would conflate convergence-speed effects with the structural effect of the projection. The uniform rate 10^{-3} was selected as the value at which the `Adam` baseline converges reliably within the epoch budget (task-1 training loss below 0.05 by epoch 10).

4.6.2 *Sub-RQ1 — Constructive Synergy or Architectural Conflict*

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ1 asks whether layering a hard geometric projection on top of a hypernetwork's soft functional regularizer yields a net benefit, or whether the momentum-stripping effect of projection methods makes a simple `Adam`-regularized hypernetwork a stronger baseline. To attribute any performance difference to the projection constraint alone — rather than to task difficulty, architecture, or training procedure — the comparison must hold across a range of difficulty levels and compression regimes. A single benchmark cannot answer this convincingly: if it is too easy, all methods reach a ceiling and no differences are observable; if it is too hard, the hypernetwork cannot learn and projection methods fail for reasons unrelated to the research question. We therefore use a three-benchmark progression, each evaluated under two panels: Panel (a), the unconstrained standalone target network, and Panel (b), the hypernetwork.

BENCHMARK 1 — SPLIT-MNIST WITH A SUFFOCATED HYPERNETWORK. A standard hypernetwork on Split-MNIST reaches near-perfect accuracy for all methods, producing a ceiling effect in which no differential effects are observable. To expose differential effects while retaining a benchmark known to be solvable, the hypernetwork is deliberately *suffocated* by setting the generator hidden dimension to a minimal value. The hidden dimension $d_h = 8$ was selected via a preliminary sweep over $d_h \in \{4, 8, 16, 32\}$ (Appendix 8). Dimensions $d_h \geq 16$ produced near-ceiling accuracy for all methods (> 0.93), leaving no measurable separation; $d_h = 4$ caused complete training failure across all methods and seeds (random-chance accuracy ≈ 0.10). $d_h = 8$ was the only configuration producing measurable separation between methods while remaining above the failure threshold for the majority of seeds. The chunk size of 64 produces 1,406 weight-generation passes per forward call (target network $\approx 90{,}000$ parameters), maximally stressing the chunk-induced gradient accumulation pathology so that any difference between normalized and unnormalized projection is also visible in this panel. Each task is trained for 15 epochs: preliminary runs showed the suffocated hypernetwork does not converge within the FOPNG paper's 5-epoch budget (task-1 loss remained above 0.20 at epoch 5 and continued decreasing through epoch 12). Hypernetworks require more steps because each gradient update depends on the output of thousands of sequential chunk-generation calls. The batch size is set to 64 rather than the canonical 10: each hypernetwork training step regenerates all 1,406 chunks before a single gradient is computed, and the larger batch reduces the step count proportionally while fitting the GPU memory budget; the method ordering is preserved regardless of batch size. Panel (a) uses the multi-head MLP standalone target with method-specific Table 1 learning rates, establishing the unconstrained reference and validating implementation correctness.

BENCHMARK 2 — SPLIT-CIFAR10 WITH A STANDARD HYPERNETWORK. Split-CIFAR10 is the primary benchmark for Sub-RQ1. Its higher input complexity creates genuine inter-task interference, preventing ceiling effects without reaching the regime where the hypernetwork fails entirely. The hypernetwork configuration — generator hidden dimension, chunk size, and embedding dimensions — is the smallest setting for which the hypernetwork achieves $\geq 70\%$ accuracy on task 1 with `Adam`, establishing sufficient capacity to learn before the projection constraint is applied; exact values are listed in Appendix 8. Each task is trained for 50 epochs, consistent with Oswald et al. (2020)'s recommendation of extended training for CIFAR-scale hypernetworks; preliminary runs confirmed 5-epoch convergence was insufficient. Panel (a) uses the FOPNG-canonical multi-head CNN with Table 1 hyperparameters.

BENCHMARK 3 — SPLIT-CIFAR100 WITH A STANDARD HYPERNETWORK. Split-CIFAR100 extends the task sequence to 10 tasks and the per-task classification problem to 10 classes, testing whether the Sub-RQ1 finding scales with sequence length and task complexity. The hypernetwork configuration matches Benchmark 2. Panel (a) uses the same CNN backbone with ten 10-class output heads. If the `Adam`-versus-projection gap grows with task count, this indicates that momentum becomes increasingly critical as the hypernetwork's task manifold grows more complex.

THREE OUTCOME PATTERNS. Comparing Panel (a) against Panel (b) within each benchmark, and across benchmarks, distinguishes three outcomes:

1. *Projection methods dominate in both panels.* The combination is constructive; Sub-RQ1 is answered affirmatively.
2. *Projection methods dominate in Panel (a) but `Adam` dominates in Panel (b).* The momentum-stripping incompatibility is the primary mechanism; Sub-RQ1 is answered negatively.
3. *Results depend on compression regime or task difficulty.* The integration is conditionally constructive, warranting finer-grained analysis of when projection adds value.

4.6.3 *Sub-RQ2 — Projection-Scoped Normalization*

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ2 asks whether the chunk-induced gradient accumulation pathology is real, characterisable, and resolvable. A single accuracy number cannot answer this: we need direct numerical evidence that the pathology occurs without normalization, that normalization resolves it, and that the resolution does not compromise projection quality.

EXPERIMENTAL DESIGN. `iFOPNG` is run on the Split-CIFAR10 hypernetwork (Benchmark 2) under three normalization conditions, all other hyperparameters held constant:

CONDITION 1 — NO NORMALIZATION (NEGATIVE CONTROL). Incoming gradient batches are appended to G without orthonormalization; the Fisher diagonal is used at its raw scale. With several hundred chunks accumulated per forward pass, the gradient columns of G and the entries of $\hat{F}$ are expected to reach extreme magnitudes, inflating the condition number of the projection kernel and driving the matrix inversion to `NaN`. This condition is the negative control that demonstrates the pathology concretely.

CONDITION 2 — GRADIENT ORTHONORMALIZATION ONLY. Each incoming gradient batch is orthonormalized via QR decomposition before insertion into G; the Fisher diagonal is used at its raw scale. This isolates whether gradient-side conditioning alone is sufficient, or whether Fisher-side scaling is also required.

CONDITION 3 — FULL NORMALIZATION (PROPOSED). Gradient batches are QR-orthonormalized at insertion, and the Fisher terms are scaled by the maximum entry of the metric tensor F_c within each projection computation. This is the proposed scheme; all other hypernetwork experiments use it.

In addition to accuracy and BWT, the condition number of the projection kernel A is logged before each inversion. A condition number that diverges in Condition 1 but remains bounded in Condition 3 constitutes direct numerical evidence for the pathology and its resolution. All conditions use the Benchmark 2 hypernetwork settings and the uniform learning rate of 10^{-3}, so any performance difference is attributable to normalization alone.

4.6.4 *Sub-RQ3 — Does iFOPNG's Elastic Preconditioner Help*

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ3 asks whether the elastic combined Fisher $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ provides a consistent retention improvement over rigid FOPNG, and whether any advantage is robust to the evaluation protocol.

BENCHMARK SELECTION. The full method suite is evaluated across the standalone benchmarks, each chosen to isolate a specific dimension of the comparison:

PERMUTED-MNIST. Domain-incremental, single shared head, high inter-task Fisher overlap. Tests long-horizon retention under input-distribution shift. If F_c accumulates historical curvature effectively, `iFOPNG` should show smaller BWT than `FOPNG` precisely where Fisher overlap is high.

SPLIT-MNIST (MULTI-HEAD, GARG ET AL. 2026). Canonical multi-head evaluation with five 2-class heads and remapped labels, replicating the conditions under which FOPNG reported its own results and providing a direct numerical comparison point.

SPLIT-MNIST (SINGLE-HEAD, Farquhar and Gal 2018). The same five binary tasks presented to a single shared 10-class head with original labels preserved. Farquhar and Gal (2018) showed that multi-head evaluation inflates apparent performance by removing output-level interference; this variant tests whether the `iFOPNG` advantage survives the harder protocol.

SPLIT-CIFAR10 AND SPLIT-CIFAR100 (MULTI-HEAD). The most discriminative standalone benchmarks: stronger inter-task interference than MNIST, clear method separation, and direct comparability with the Panel (a) references of Sub-RQ1. Split-CIFAR100 additionally tests whether the elastic advantage holds as the sequence extends to 10 tasks.

WHAT A DIFFERENCE REVEALS. Comparing `iFOPNG` against `FOPNG` across all benchmarks tests whether the elastic preconditioner provides a consistent, robust improvement or a benchmark-specific artefact. Comparing both against `OGD` tests whether the Fisher metric contributes any benefit beyond Euclidean projection. Comparing the two Split-MNIST variants tests the protocol-robustness claim of Farquhar and Gal (2018) directly. All hyperparameters for these experiments follow Garg et al. (2026) Table 1 as described in Section 4.6.1; no tuning was performed.

4.6.5 *Sub-RQ4 — Fisher Accumulation Operator*

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ4 holds the elastic formulation of Sub-RQ3 fixed and isolates the operator used to accumulate the historical Fisher F_{old}. The non-decaying maximum $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ guarantees that any parameter ever deemed important to a past task retains that importance permanently. This is protective for early tasks but monotonic by construction: the elastic inertia boundary can only tighten, never relax, so in a sufficiently long sequence the accumulated metric may immobilise the parameters needed to learn later tasks.

EXPERIMENTAL DESIGN. `iFOPNG` is run under two accumulation operators, all other settings held constant:

MAXIMUM (PROPOSED). $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ — non-decaying.

EXPONENTIAL MOVING AVERAGE. $F_{\text{old}} \leftarrow (1-\alpha)\, F_{\text{old}} + \alpha\, F_{\text{new}}$ — decaying, allowing historical importance to fade.

The comparison is run on the longest available task sequences — Permuted-MNIST and Split-CIFAR100 — since the plasticity cost of monotonic accumulation, if present, only becomes observable as the sequence length grows. Per-task accuracy is tracked across the full sequence rather than reported only as a final aggregate: the diagnostic signature of plasticity paralysis is a progressive decline in *newly learned* task accuracy ($a_{t,t}$ falling as t increases), distinct from forgetting, which appears as decline in *previously learned* task accuracy. If the maximum operator shows superior early-task retention but degrading $a_{t,t}$ in late tasks, while the moving average shows the opposite trade, the two failure modes are cleanly separated and Sub-RQ4 is answered in full.

5 RESULTS

This section reports the empirical outcomes of the four sub-research questions in the order stated in the introduction. For each, we describe the experimental scope, present the key quantitative findings, and highlight the most important patterns; detailed interpretation is reserved for the Discussion. *Average accuracy* (A_T) is the primary evaluation metric, and *backward transfer* (BWT) is reported as a secondary measure of forgetting. Standalone target-network results that establish the reference behaviour of projection methods in their original, non-hypernetwork setting are provided in Appendix B (Figures 7–9).

5.1 *Sub-RQ1: Constructive Synergy or Architectural Conflict?*

This subsection examines whether combining gradient projection with the hypernetwork's functional regularizer yields better continual learning than either component alone. Three hypernetwork regimes are evaluated: a standard-capacity Split-CIFAR10 network, an extremely compressed Split-MNIST network (suffocated regime, $d_h = 8$), and a ten-task Split-CIFAR100 network.

STANDARD HYPERNETWORK — SPLIT-CIFAR10. Figure 1 presents average accuracy and BWT for all eight methods on Split-CIFAR10 with a standard-capacity hypernetwork and full normalization enabled (see Section 5.2). Plain `SGD` at 84.1% establishes the unconstrained lower reference point, and `Adam` at 90.1% defines the upper frontier a standalone hypernetwork can reach without projection. Every projection method surpasses SGD: `iFOPNG` leads at 85.6%, followed by `EWC` (81.5%), `FNG` (80.9%), `OGD` (78.3%), `FOPNG` (77.5%), and `ONG` (67.2%), confirming a partial constructive synergy with the hypernetwork's regularizer. However, none match `Adam`, which leads by 4.5 pp over the best projection method.

The BWT values reveal an instructive asymmetry: `Adam` has effectively zero backward transfer (BWT ≈ 0.000), while all projection methods exhibit small but nonzero forgetting (iFOPNG: -0.013; FOPNG: -0.015).

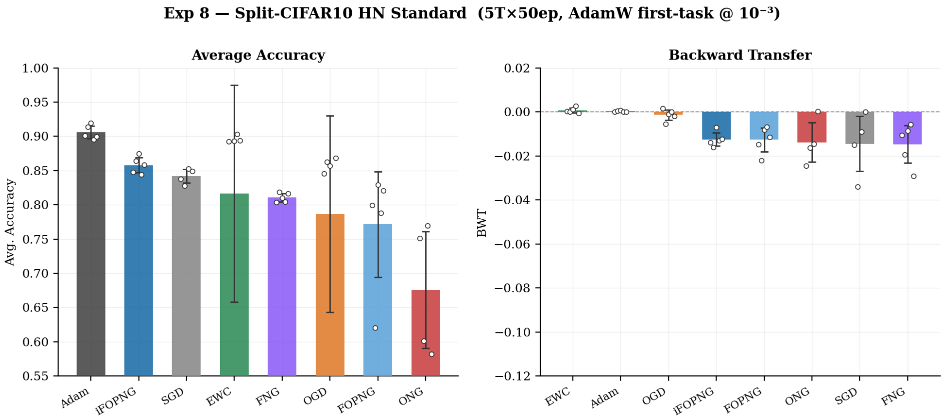

Figure 1: **Split-CIFAR10 hypernetwork (5 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3}).** Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. Methods are sorted by average accuracy on the left panel. `Adam` defines the projection-free upper frontier; `SGD` defines the unconstrained lower reference. All projection methods trained with full normalization (Section 5.2) exceed SGD but fall short of Adam.

SUFFOCATED REGIME — SPLIT-MNIST, $d_h = 8$. The embedding dimension $d_h = 8$ was selected via a three-point sweep over $d_h \in \{4, 8, 16\}$, presented as an ablation in Figure **??**. That sweep shows the projection–`Adam` gap is largest at $d_h = 8$ (23.5 pp), falls to 14.0 pp at $d_h = 4$ where seed variance becomes extreme and results near-random, and shrinks to 9.4 pp at $d_h = 16$ as the projection methods partially recover. $d_h = 8$ was retained as the most diagnostically informative level: the failure mode is severe and replicable, but the network has sufficient capacity for the projections to converge consistently.

Figure 2 presents the full method comparison at $d_h = 8$. `Adam` leads at 93.4%. The Fisher-free projection method `OGD` is the strongest projection baseline at 86.6%, followed by `EWC` at 84.5% and `SGD` at 79.8%. The Fisher-based methods fall sharply below: `iFOPNG` at 69.9%, `FOPNG` at 59.9%, and `FNG` at 54.8% — gaps of 23–39 pp against `Adam`.

The BWT panel reveals the mechanism underlying this collapse. `iFOPNG`, `FOPNG`, and `FNG` all achieve near-zero backward transfer (BWT ≈ 0.00), while `Adam` has small negative BWT (-0.015) and `SGD` forgets more substantially (-0.13). The Fisher-based projection methods are not failing at retention — they are failing at plasticity. In a $d_h = 8$ bottleneck, the curvature-weighted projection constraint exhausts the few available gradient directions protecting prior tasks, leaving negligible residual capacity for learning new ones. `OGD`'s relative resilience is consistent with this interpretation: as a Fisher-free method its subspace constraint is less aggressive, leaving more gradient budget for forward transfer. Seed-level variance is substantially elevated for all projection methods, and several methods completed fewer than three seeds at this compression level; confidence intervals should be interpreted cautiously.

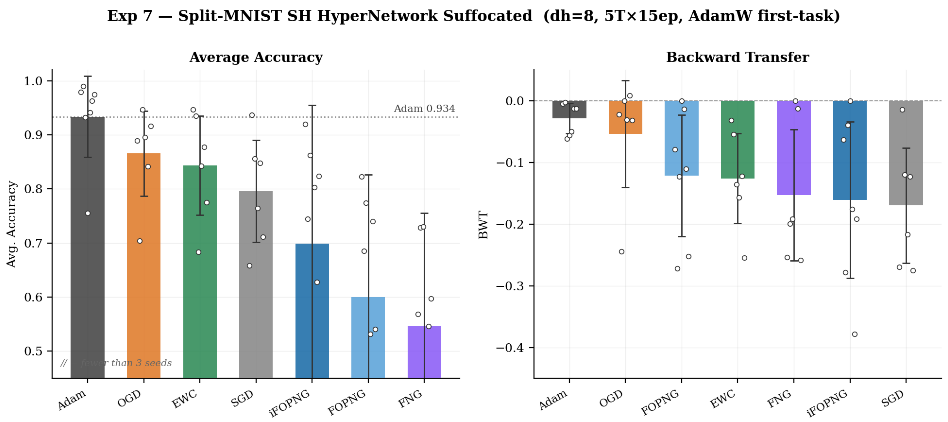

Figure 2: **Split-MNIST SH hypernetwork, suffocated regime ($d_h = 8$, 5 tasks, 15 epochs per task, `AdamW` first-task initialisation).** Left: average accuracy (bars show mean over seeds; circles show individual seeds; note that several methods completed fewer than three seeds). Right: backward transfer. The dotted line marks `Adam`'s mean accuracy (93.4%). Fisher-based projection methods (`iFOPNG`, `FOPNG`, `FNG`) achieve near-zero BWT but much lower accuracy than `Adam`, revealing a plasticity failure rather than a retention failure in the extreme compression regime. The Fisher-free `OGD` is the best-performing projection method at 86.6%.

EXTENDED TASK COUNT — SPLIT-CIFAR100. At ten tasks on Split-CIFAR100 (Appendix Figure 11), `EWC` unexpectedly leads at 45.5%, with `Adam` second at 42.0% and `iFOPNG` third at 37.8% (random baseline: 10.0%). This partial rank reversal, in which soft regularization overtakes both Adam and gradient projection, differs from the five-task CIFAR10 pattern and is addressed in the Discussion.

5.2 *Sub-RQ2: Normalization Resolves the Gradient Pathology*

This subsection reports the three-condition normalization ablation (Split-CIFAR10 hypernetwork; 197 chunks, ≈6000 output neurons; unnormalized conditions ran for 10 epochs per task with early stopping, full normalization for 50 epochs). Figure 3 traces the $\log_{10}$ condition number of the projection kernel $G^\top \hat{F} G$ over training steps; Figure 4 shows the resulting average accuracy and BWT under each condition.

WITHOUT NORMALIZATION. The right panel of Figure 3 shows the condition number reaching $10^{10.8}$ at the T2$\to$T3 task transition for the most-affected seed, stabilizing near $10^{8.8}$–$10^{9.0}$ for two others. At this scale, numerical inversion of the projection kernel is unreliable, effectively zeroing the projected gradient so that the method operates as an unconstrained optimizer with unstable Fisher scaling. The result is 72.2% average accuracy and BWT $= -0.070$ — worse than plain SGD (84.1%) despite identical hyperparameters. The pathology therefore not only eliminates the benefit of projection but actively degrades performance below the projection-free baseline.

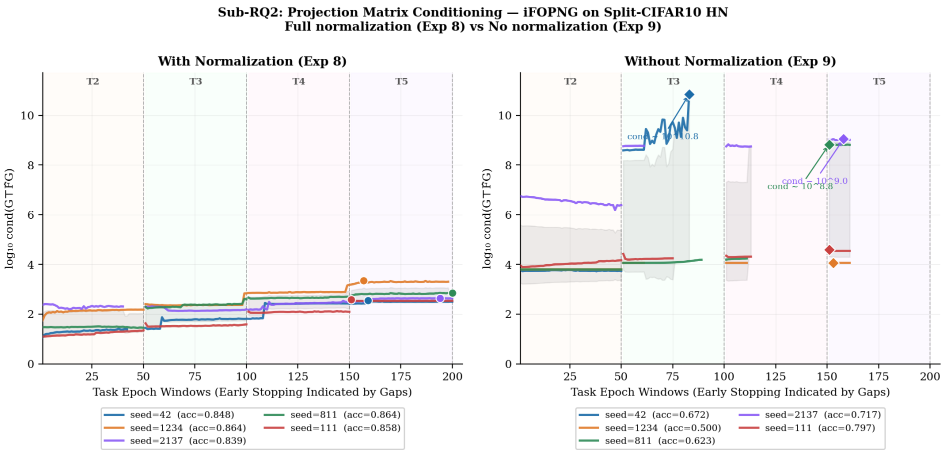

Figure 3: **Projection kernel condition number over training — `iFOPNG` on Split-CIFAR10 HN.** Each line is one seed (legend shows final average accuracy). Task boundaries are marked by vertical dashed lines; gaps indicate early stopping. Left (full normalization): $\log_{10} \kappa(G^\top \hat{F} G)$ remains bounded within $[10^{1.3}, 10^{3.3}]$ throughout. Right (no normalization): condition number spikes to $10^{10.8}$ at the T2$\to$T3 transition, causing projection failure. Diamond markers indicate end-of-task condition number values.

GRADIENT-ONLY NORMALIZATION. Applying unit-ℓ_2 column normalization to G alone, without rescaling $\hat{F}$, fails to resolve the pathology. Average accuracy falls further to 68.9% (BWT $= -0.060$) and variance across seeds remains high (Figure 4). As the left panel of Figure 3 illustrates, the raw Fisher magnitudes from the chunked backward passes still span several orders of magnitude, preserving the ill-conditioning of $G^\top \hat{F} G$ even when G itself is normalized. Partial normalization is therefore insufficient.

FULL NORMALIZATION. Combining column-wise ℓ_2 normalization of G with absolute-maximum scaling of $\hat{F}$ fully resolves the pathology. The left panel of Figure 3 shows $\log_{10} \kappa(G^\top \hat{F} G)$ bounded within $[10^{1.3}, 10^{3.3}]$ across all seeds and task transitions — a reduction of more than seven orders of magnitude. Figure 4 shows the result: average accuracy recovers to 85.8% and BWT improves to -0.013, a 5.7 pp reduction in forgetting relative to no normalization. The improvement is consistent across all three seeds. Full normalization is adopted for all hypernetwork experiments in this thesis.

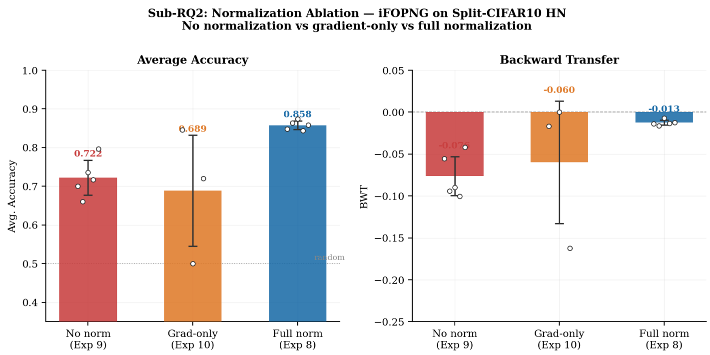

Figure 4: **Normalization ablation — `iFOPNG` on Split-CIFAR10 HN (Sub-RQ2).** Three conditions: no normalization, gradient-only normalization, and full normalization. Left: average accuracy with mean annotated on each bar. Right: backward transfer. Circles show individual seeds; bars show the mean. The dotted line on the left panel marks random-chance performance (50% for binary per-task classification).

DIAGNOSTIC: SGD FIRST-TASK CONVERGENCE THRESHOLD. A diagnostic experiment (Appendix Figure 12) identifies an inconsistency in the FOPNG paper's stated first-task protocol. At FOPNG's learning rate of $\eta = 10^{-5}$ on Split-MNIST, five epochs of SGD training yield only $\approx 21\%$ accuracy — barely above the 50% random baseline. The empirical convergence threshold lies between 5×10^{-5} and 10^{-4}. All standalone experiments in this thesis use $\eta = 10^{-3}$ for first-task SGD, deviating from but empirically improving upon the original protocol.

5.3 *Sub-RQ3: Parameter Inertia via the Combined Fisher Metric*

This subsection evaluates whether replacing FOPNG's task-specific natural gradient metric $\hat{F}_{\text{new}}$ with the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ systematically improves task retention across benchmarks. Figure 5 presents mean average accuracy for both methods across all six benchmarks.

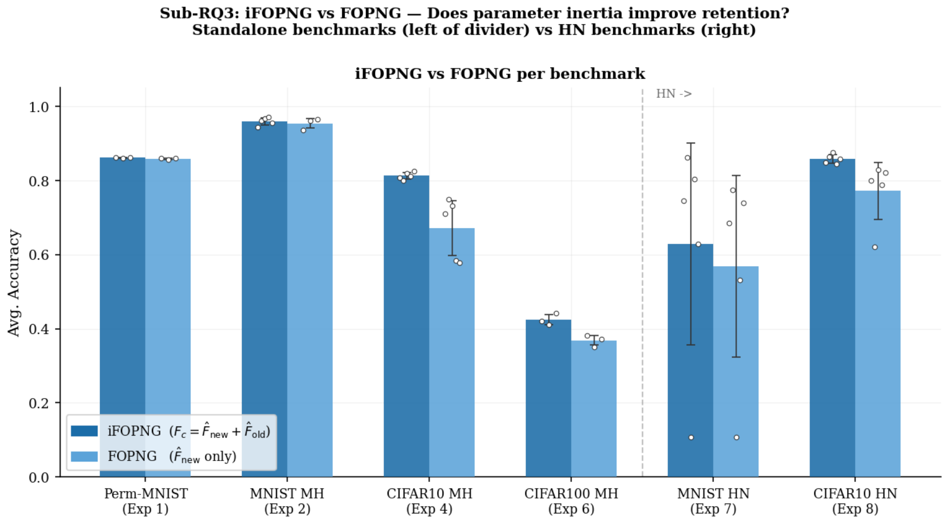

Figure 5: **iFOPNG vs. FOPNG across six benchmarks (Sub-RQ3).** Grouped bars show mean average accuracy; circles show individual seeds; error bars show ± 1 standard deviation. The dashed vertical line separates standalone (left) from hypernetwork (right) settings. `iFOPNG` uses the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$; `FOPNG` uses only $\hat{F}_{\text{new}}$.

On the two simplest benchmarks — Permuted-MNIST and Split-MNIST multi-head — the two methods are nearly indistinguishable (86.0% vs. 85.9% and 95.6% vs. 95.2% respectively). Both tasks are low-dimensional with limited inter-task Fisher overlap, so $\hat{F}_{\text{old}}$ contributes little additional mass beyond what $\hat{F}_{\text{new}}$ already captures; parameter inertia has no accumulated weight to draw on.

A clear separation emerges on harder benchmarks. On Split-CIFAR10 multi-head, `iFOPNG` reaches 81.1% versus FOPNG's 67.4%, a gap of 14.1 pp. On Split-CIFAR100 multi-head the advantage is 5.6 pp (42.7% vs. 37.1%).

The advantage carries over into the hypernetwork settings: 9.8 pp on Split-MNIST HN (70.0% vs. 60.0%) and 8.6 pp on Split-CIFAR10 HN (85.6% vs. 77.5%). Across all six benchmarks the advantage of `iFOPNG` grows with task difficulty and cumulative Fisher overlap.

5.4 *Sub-RQ4: Fisher Accumulation Operator*

This subsection compares two strategies for accumulating F_{old} across tasks under the iFOPNG combined Fisher metric: the element-wise maximum operator (MAX) and an exponential moving average (EMA). Both are evaluated on a twenty-task Permuted-MNIST sequence with five random seeds. Figure 6 presents the average accuracy and backward transfer trajectories over all twenty tasks; the per-seed statistical comparison is provided in Appendix Figure 13.

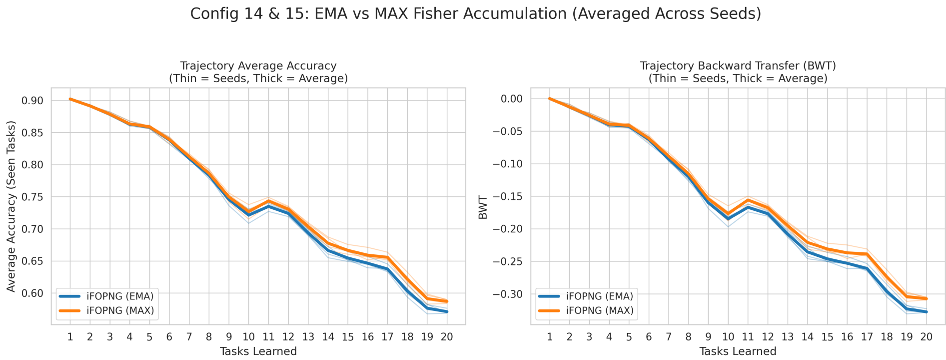

Figure 6: **EMA vs. MAX Fisher accumulation trajectories on 20-task Permuted-MNIST (Sub-RQ4),** $n = 5$ **seeds.** Thick lines show the seed-averaged trajectory; thin lines show individual seeds. Left: average accuracy over tasks seen so far, evaluated after each task is trained. Right: backward transfer over the same sequence. MAX leads EMA by a consistent margin across all five seeds; both strategies produce closely tracking trajectories throughout the full twenty-task sequence.

A paired t-test on final average accuracy establishes MAX as the superior operator: $t(4) = 14.43$, $p < 0.001$, Cohen's $d = 7.22$, with MAX leading by a consistent 1.51–1.65 pp across every seed pair without exception (Appendix Figure 13). The result is statistically unambiguous and replicates across all five seeds. In absolute terms the margin is practically negligible — 1.60 pp on a benchmark where both variants decline by more than 30 pp over the full sequence — but the direction is consistent and the effect size large, leaving no ambiguity about which operator to prefer.

The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

6 DISCUSSION

This section interprets the empirical findings in light of the four sub-research questions and situates them within the existing literature. The overarching goal was to determine whether gradient projection methods can be fruitfully integrated into chunked hypernetworks for continual learning, and to identify the structural conditions under which this integration succeeds or fails.

6.1 *Sub-RQ1: Constructive Synergy or Architectural Conflict?*

Projection methods do not outperform a well-tuned regularized hypernetwork optimized via `Adam`. The gap of 4.5 pp at task 5 in favor of `Adam` is consistent across seeds and widens substantially in the suffocated regime ($d_h = 8$, Split-MNIST SH), where it reaches 20–40 pp. We attribute this to a fundamental architectural incompatibility. Projection methods operate by modifying the raw gradient before the optimizer state update, which means the first-order momentum estimate and the adaptive scaling factor maintained by Adam are effectively discarded at each projected step. In the highly nonlinear, compressed weight space of a hypernetwork meta-generator, momentum is not merely a convergence accelerant — it is a structural regularizer that prevents the optimizer from committing prematurely to poorly-conditioned gradient directions. Re-initializing the optimizer's internal state at each step destabilizes the generative manifold and reduces the quality of the generated weights, particularly under extreme bottleneck compression. This finding is consistent with Hu et al. (2026), who demonstrate that gradient modification methods interact adversely with Adam's internal state under certain curvature regimes, and aligns with the broader observation that projection-based continual learning methods were originally designed for SGD-like updates.

The ten-task Split-CIFAR100 result introduces a partial rank reversal that complicates the Sub-RQ1 conclusion: `EWC` unexpectedly leads at 45.5%, overtaking both `Adam` (42.0%) and `iFOPNG` (37.8%). This inversion suggests that at higher task counts, soft functional regularization in output space may dominate hard gradient projection because the projection memory cannot hold sufficient historical directions within a fixed column budget. As the number of tasks grows, the gradient subspace spanned by G becomes an increasingly coarse approximation to the full historical interference manifold, while the functional regularizer in output space grows more informative with each additional stored weight snapshot. Whether this effect persists at the twenty-task scale, and whether an expanded memory budget would recover the projection advantage, remains to be investigated.

FIRST-TASK INITIALIZATION AND GRADIENT CONTAMINATION. The comparison between `AdamW` and `Adam` first-task initialization confirmed the contamination hypothesis for gradient projection methods in the hypernetwork setting. Standard `Adam` couples weight decay into the gradient estimate as an additive ℓ_2 term, so both the gradient memory G and the Fisher diagonal $\hat{F}$ collected at the end of task 1 inherit a component aligned with the weight magnitude vector rather than the task loss curvature. This contaminates the projection subspace: downstream tasks are projected away from directions spuriously marked as historical but actually reflecting optimizer regularization rather than task geometry. `AdamW` initialization, which applies weight decay as a decoupled parameter-shrinkage step, consistently produced lower first-task trajectory variance and a cleaner initial projection basis. This finding carries a practically important implication for any system bootstrapping a projection memory from first-task gradients: the decoupled optimizer should be preferred at that stage regardless of which optimizer is used in subsequent tasks.

6.2 *Sub-RQ2: Projection-Scoped Normalization Resolves the Pathology*

This result contributes a practical engineering solution that is necessary for any future work combining hard projection constraints with chunked meta-learners, regardless of which projection algebra is employed. The normalization is applied locally within the projection computation and does not alter the underlying optimization landscape: subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation changes the geometry of the projection. The fundamental problem is therefore purely numerical rather than conceptual: the intersection of chunked hypernetworks and gradient projection is geometrically coherent but arithmetically fragile without explicit local rescaling.

6.3 *Sub-RQ3: Parameter Inertia via the Elastic Fisher Preconditioner*

The `iFOPNG` advantage scales monotonically with benchmark difficulty across all six settings (Figure 5): negligible where inter-task Fisher overlap is low, substantial on high-overlap benchmarks. F_c inflates the effective curvature of historically important coordinates, suppressing over-aggressive natural gradient steps without any penalty term or stored reference point.

This mechanism stands in direct contrast to EWC, which achieves a similar dampening effect via an explicit quadratic penalty pulling parameters toward a reference point. `iFOPNG` applies no such restoring force and maintains no reference: a parameter that mattered before is not pulled back, it is made harder to displace. The empirical advantage of `iFOPNG` over `FOPNG` on high-overlap benchmarks confirms that the rigidity of the pure projection boundary is a genuine limitation of FOPNG, and that lifting it via the combined metric F_c is a principled and effective remedy. Nevertheless, `iFOPNG` does not close the gap to `Adam`, confirming that the elastic preconditioner addresses rigidity but cannot recover the momentum mechanism that `Adam` leverages for efficient navigation of the generative manifold.

6.4 *Sub-RQ4: Fisher Accumulation Operator*

This null result is itself informative. The theoretical argument for EMA was that monotonic MAX growth would impose a steadily increasing effective mass on parameters that were important to any prior task, eventually over-constraining the natural gradient and harming plasticity on new tasks. The absence of this effect on Permuted-MNIST suggests that, under the normalization scheme applied here (which rescales both Fisher terms by the current maximum of F_c), the absolute scale of F_{old} does not enter the projection computation — only its relative distribution does. The max-normalized Fisher is therefore insensitive to whether the accumulation operator is decaying or non-decaying, as long as the distribution of importance across parameters is preserved.

Two limitations of this finding should be noted. First, Permuted-MNIST is a structurally homogeneous benchmark in which all tasks share the same label space and architecture; inter-task Fisher overlap is low and approximately uniform across tasks. On benchmarks with heterogeneous task distributions and non-uniform overlap, the two operators may diverge as the historical Fisher accumulates disproportionate mass in task-specific parameter subspaces. Second, the experiment used a fixed EMA decay of $\alpha = 0.5$; an adaptive schedule that increases decay as the task sequence grows longer may produce a more favorable comparison for EMA under long-horizon retention.

6.5 *Limitations*

Several limitations should be acknowledged.

First, the hypernetwork results presented here are restricted to Split-MNIST and Split-CIFAR10 with five tasks. Longer task sequences may produce different relative orderings: in particular, the advantage of hard projection constraints over soft functional regularization may grow with sequence length, as soft penalties are known to suffer from compounding output drift (Farquhar & Gal, 2018), while the projection subspace scales with the gradient budget rather than the task count. The preliminary ten-task Split-CIFAR100 result, in which EWC overtakes both `Adam` and `iFOPNG`, is suggestive but insufficient to establish a trend; a systematic sweep over task count is warranted.

Second, the hypernetwork architecture used in this study is intentionally shallow, with small hidden dimensions ($d_h \in \{8, 16\}$) and embedding dimensions of 32. Larger hypernetworks, such as those used for language model weight generation (Chauhan, Zhou, Lu, Molaei, & Clifton, 2024), may exhibit qualitatively different gradient accumulation dynamics —

both in terms of the severity of the normalization pathology and in the relative contribution of momentum to meta-generator stability. The suffocated regime ($d_h = 8$) explored here provides an empirical lower bound on expressivity, but an upper bound in a high-capacity regime remains uncharted.

Third, the proposed normalization scheme — QR orthonormalization of the gradient basis combined with absolute-maximum scaling of the metric tensor F_c — is heuristic in nature. It discards the eigenvalue structure of the pullback metric and applies isotropic rather than curvature-aware rescaling to the chunk embedding directions. The principled alternative is the Riemannian gradient under the pullback metric $G_i = J_i^\top J_i$, where $J_i = \partial W_i / \partial c_i$ is the Jacobian of the chunk-generating map:

$$\Delta c_i = -\eta \, (J_i^\top J_i + \lambda I)^{-1} \nabla_{c_i} \mathcal{L}. \tag{19}$$

This preconditioner was implemented using batched forward-mode automatic differentiation across all K chunks simultaneously. When applied to the raw gradient *prior* to the FOPNG projection step, however, it consistently degraded continual learning performance in preliminary evaluation: backward transfer worsened by approximately 0.33 units on Split-MNIST SH relative to the heuristic baseline. We attribute this to a metric mismatch: the FOPNG projection operator is calibrated for gradients in Euclidean parameter space, and preconditioning only the chunk embedding component before projection produces a geometrically inconsistent input to the kernel $G^\top \hat{F} G$. The principled resolution — applying the natural gradient to the projected update v^* rather than the raw gradient, so that the projection sees Euclidean inputs and the Riemannian step follows — is left to future work.

7 CONCLUSION

This thesis investigated the extent to which gradient projection methods can be successfully integrated into chunked hypernetwork architectures for continual learning, and identified the structural conditions under which that integration yields benefit or harm. The primary research question decomposed into four sub-questions addressing constructive synergy, gradient normalization, the elastic Fisher metric, and the Fisher accumulation operator.

SUB-RQ1 — PARTIAL SYNERGY BOUNDED BY THE MOMENTUM PROBLEM. The integration is constructive but incomplete. No projection variant matches a standalone regularized hypernetwork optimized via `Adam`, which leads by approximately 4.5 pp on Split-CIFAR10 and by 20–

40 pp in the extreme compression regime ($d_h = 8$). The primary obstacle is structural: projection methods modify the raw gradient before the optimizer state update, effectively discarding the first-order momentum that stabilizes generative manifold navigation in compressed weight spaces. This interaction has been independently identified as problematic under certain curvature regimes (Hu et al., 2026), and resolving it will require projection-compatible adaptive optimizers rather than post-hoc gradient modification.

SUB-RQ2 — NORMALIZATION IS A NECESSARY CONDITION. Without explicit local rescaling, projection methods fail catastrophically on chunked hypernetworks: the condition number of the projection kernel reached $10^{10.8}$, rendering matrix inversion numerically unreliable. QR orthonormalization of the gradient basis combined with absolute-maximum scaling of the metric tensor F_c reduces the condition number by more than seven orders of magnitude — bounding it within $[10^{1.3}, 10^{3.3}]$ — and fully recovers projection performance across all three seeds on the Split-CIFAR10 hypernetwork. This normalization scheme is a necessary engineering precondition for any future combination of gradient projection with chunked meta-learners, regardless of which projection algebra is employed.

SUB-RQ3 — PARAMETER INERTIA IMPROVES RETENTION ON HIGH--OVERLAP BENCHMARKS. Replacing `FOPNG`'s task-specific Fisher metric with the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ consistently improves task retention when inter-task Fisher overlap is substantial. The advantage of `iFOPNG` over `FOPNG` grows monotonically with benchmark difficulty: negligible on Permuted-MNIST (+0.1 pp), substantial on Split-CIFAR10 multi-head (+14.1 pp), and meaningful in the hypernetwork settings (+9.8 pp and +8.6 pp). `iFOPNG` thereby provides a principled and practically effective improvement over `FOPNG` at no additional hyperparameter cost.

SUB-RQ4 — MAX ACCUMULATION IS THE CLEAR PRACTICAL CHOICE. On a twenty-task Permuted-MNIST sequence evaluated across five seeds, the element-wise maximum (MAX) and exponential moving average (EMA) accumulation strategies produced nearly parallel accuracy and backward transfer trajectories throughout the full sequence. A paired t-test confirms that MAX is the superior operator: $t(4) = 14.43$, $p < 0.001$, Cohen's $d = 7.22$, with MAX leading by a consistent 1.51–1.65 pp across every seed pair. The effect is systematic and statistically unambiguous, though small in absolute terms: 1.60 pp on a benchmark where both strategies decline by more than 30 pp over the full sequence — practically negligible in deployment terms. The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

IMPLICATIONS FOR THE FIELD. The results establish a clear empirical boundary for the current generation of gradient projection methods: they can be made numerically stable in chunked hypernetwork settings via local rescaling, and their Fisher geometry can be improved via elastic accumulation, but the momentum incompatibility prevents them from reaching the performance frontier of adaptive unconstrained optimizers in generative parameter spaces. The normalization contribution is directly applicable to any method that maintains a gradient subspace memory alongside a chunked weight generator. The iFOPNG contribution extends naturally to any continual learning setting where Fisher overlap accumulates across tasks. Together, they narrow the gap between projection-based and regularization-based approaches, and provide a principled foundation for the next step: a projected adaptive optimizer that respects both the geometric constraints of the projection and the curvature structure that adaptive moment estimates capture.

FUTURE DIRECTIONS. Three directions are most directly motivated by the findings of this thesis. First, a projected variant of Adam that maintains momentum across projected steps — rather than discarding it at each update — would address the primary mechanism identified in Sub-RQ1 and has been partially explored in concurrent work (Hu et al., 2026). Second, the natural gradient preconditioner for chunk embeddings under the pullback metric $G_i = J_i^\top J_i$ was implemented and found to interact destructively with the FOPNG projection when applied to the raw gradient; applying it to the projected update v^* instead would resolve the metric mismatch and merits empirical evaluation. Third, extending the evaluation to longer task sequences and larger hypernetwork capacities would test whether the partial synergy identified here scales favourably as the projection subspace budget grows relative to the task manifold — the regime in which hard constraints are expected, by theory, to provide the greatest benefit over soft regularization.

ACKNOWLEDGEMENTS

The author thanks dr. Giacomo Spiegler for supervision and valuable feedback throughout this project.

REFERENCES

Amari, S.-i. (n.d.). Natural gradient works efficiently in learning. , *10*(2), 251–276. Retrieved 2026-05-31, from https://ieeexplore.ieee.org/document/6790500 doi: 10.1162/089976698300017746

Bennani, M. A., Doan, T., & Sugiyama, M. (2020). *Generalisation guarantees for continual learning with orthogonal gradient descent* (No. arXiv:2006.11942). arXiv. Retrieved 2026-04-22, from http://arxiv.org/abs/2006.11942 doi: 10.48550/arXiv.2006.11942

Chauhan, V. K., Zhou, J., Lu, P., Molaei, S., & Clifton, D. A. (2024). A brief review of hypernetworks in deep learning. , *57*(9), 250. Retrieved 2026-02-21, from http://arxiv.org/abs/2306.06955 doi: 10.1007/s10462-024-10862-8

De Lange, M., Aljundi, R., Masana, M., Parisot, S., Jia, X., Leonardis, A., ... Tuytelaars, T. (2022). A continual learning survey: Defying forgetting in classification tasks. , *44*(7), 3366–3385. Retrieved 2026-05-16, from https://ieeexplore.ieee.org/document/9349197/ doi: 10.1109/TPAMI.2021.3057446

Farajtabar, M., Azizan, N., Mott, A., & Li, A. (2020). *Orthogonal gradient descent for continual learning* (No. arXiv:1910.07104). arXiv. Retrieved 2026-02-23, from http://arxiv.org/abs/1910.07104 doi: 10.48550/arXiv.1910.07104

Farquhar, S., & Gal, Y. (2018). *Towards robust evaluations of continual learning* (No. arXiv:1805.09733). arXiv. Retrieved 2026-04-05, from http://arxiv.org/abs/1805.09733 doi: 10.48550/arXiv.1805.09733

Fuente, N. D. L., Pilligua, M., Vidal, D., Soutiff, A., Curreli, C., Cremers, D., & Barsky, A. (2025). *Prototype augmented hypernetworks for continual learning* (No. arXiv:2505.07450). arXiv. Retrieved 2026-02-28, from http://arxiv.org/abs/2505.07450 doi: 10.48550/arXiv.2505.07450

Garg, I., Kolhe, N., Peng, A., & Gopalam, R. (2026). *Fisher-orthogonal projected natural gradient descent for continual learning* (No. arXiv:2601.12816). arXiv. Retrieved 2026-02-21, from http://arxiv.org/abs/2601.12816 doi: 10.48550/arXiv.2601.12816

Grossberg, S. (1980). How does a brain build a cognitive code? , *87*(1), 1–51. (Place: US Publisher: American Psychological Association) doi: 10.1037/0033-295X.87.1.1

Ha, D., Dai, A., & Le, Q. V. (2016). *HyperNetworks* (No. arXiv:1609.09106). arXiv. Retrieved 2026-02-21, from http://arxiv.org/abs/1609.09106 doi: 10.48550/arXiv.1609.09106

Hemati, H., Lomonaco, V., Bacciu, D., & Borth, D. (2023). *Partial hypernetworks for continual learning* (No. arXiv:2306.10724). arXiv. Retrieved 2026-02-28, from http://arxiv.org/abs/2306.10724 doi: 10.48550/arXiv.2306.10724

Hu, Y., Yu, Z., Cheng, Z., Liu, W., & Song, L. (2026). *Hidden failure modes of gradient modification under adam in continual learning, and adaptive decoupled moment routing as a repair* (No. arXiv:2604.22407). arXiv. Retrieved 2026-05-16, from http://arxiv.org/abs/2604.22407 doi: 10.48550/arXiv.2604.22407

Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., ... Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. , *114*(13), 3521–3526. Retrieved 2026-02-23, from http://arxiv.org/abs/1612.00796 doi: 10.1073/pnas.1611835114

McCloskey, M., & Cohen, N. J. (1989). *Catastrophic interference in connectionist networks: The sequential learning problem* (Vol. 24). Academic Press. Retrieved from https://www.sciencedirect.com/science/article/pii/S0079742108605368 doi: https://doi.org/10.1016/S0079-7421(08)60536-8

Oswald, J. v., Henning, C., Grewe, B. F., & Sacramento, J. (2020). *Continual learning with hypernetworks* (No. arXiv:1906.00695). arXiv. Retrieved 2026-02-21, from http://arxiv.org/abs/1906.00695 doi: 10.48550/arXiv.1906.00695

8 APPENDIX

APPENDIX A: IFOPNG THEOREM AND PROOF

Notation

Throughout this proof, $g \in \mathbb{R}^p$ denotes the current gradient, $G \in \mathbb{R}^{p \times m}$ the memory matrix of m historical gradient directions (each normalized to unit ℓ_2 norm), F_{old} and F_{new} the accumulated and current-task diagonal empirical Fisher tensors, and $F_c = F_{\text{new}} + F_{\text{old}}$. We treat diagonal Fisher matrices as vectors acting by element-wise multiplication; all matrix operations below are defined accordingly.

Assumption 1 (Regularity). *F_c is positive definite and the matrix $A = G^\top F_{old} F_c^{-1} F_{old} G$ is invertible.*

Theorem 1 (iFOPNG Update). *Under Assumption 1, the solution to*

$$\begin{aligned} &\min_u \; \|g - F_c^{-1} u\|_{F_c}^2 \\ s.t. \quad & F_{old}^{-1} u \in \mathrm{Col}(G), \\ & \|v\|_{F_c} = \eta, \end{aligned} \tag{20}$$

with parameter update $v = c\, F_c^{-1}(g - u)$, is given by (15), where $P = I - F_{old}\, G\, A^{-1}\, G^\top F_{old}$.

Proof. **Step 1.** The constraint $F_{\text{old}}^{-1} u \in \mathrm{Col}(G)$ implies $u = F_{\text{old}}\, G\, \alpha$ for some $\alpha \in \mathbb{R}^m$.

Step 2. Substituting and expanding using $\|w\|_{F_c}^2 = w^\top F_c\, w$:

$$\|g - F_c^{-1} u\|_{F_c}^2 = g^\top F_c\, g - 2\, g^\top F_{\text{old}}\, G\, \alpha + \alpha^\top A\, \alpha. \tag{21}$$

Step 3. Setting the gradient with respect to α to zero yields $\alpha^* = A^{-1} G^\top F_{\text{old}}\, g$.

Step 4. Substituting back: $u^* = F_{\text{old}}\, G\, A^{-1}\, G^\top F_{\text{old}}\, g$, so $g - u^* = Pg$.

Step 5. Applying the trust-region constraint $\|v\|_{F_c} = \eta$ to $v = c\, F_c^{-1} P g$ gives $c = \eta / \sqrt{(Pg)^\top F_c^{-1} P g}$, yielding the stated update. □

APPENDIX B: STANDALONE TARGET-NETWORK RESULTS

This appendix presents the full method comparison on standalone target networks — configurations where a conventional MLP is trained directly via gradient projection, without any hypernetwork wrapper or functional regularizer. These results establish the baseline behaviour of each projection method in its intended setting, prior to the hypernetwork integration examined in the main text. Four benchmark streams are reported: Permuted-MNIST (5 tasks) and Split-MNIST multi-head (5 tasks) in Figure 7, Split-CIFAR10 multi-head (5 tasks) in Figure 8, and Split-CIFAR100 multi-head (10 tasks) in Figure 9. Bars show mean accuracy over seeds; circles show individual seeds.

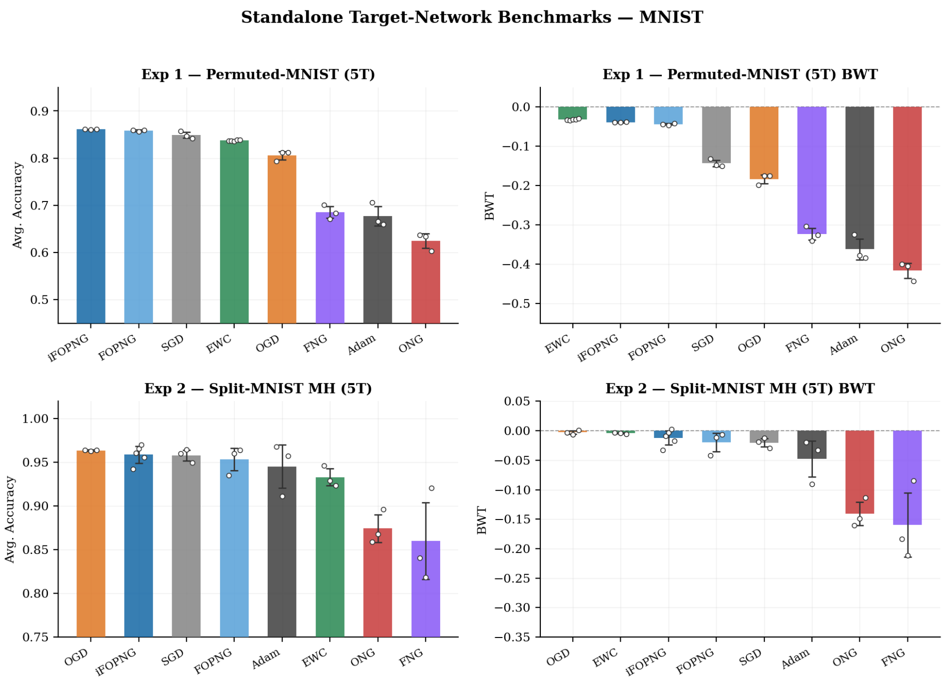

Figure 7: **Standalone target-network results — Permuted-MNIST (5 tasks, top row) and Split-MNIST multi-head (5 tasks, bottom row).** Left column: average accuracy. Right column: backward transfer. Bars show mean over seeds; circles show individual seeds. *Permuted-MNIST.* `iFOPNG` and `FOPNG` lead at 86.1% and 85.8% respectively with near-zero backward transfer, closely followed by `SGD` (84.9%) and `EWC` (83.8%). `Adam` (68.2%) and `ONG` (62.6%) exhibit substantially negative BWT (≈ -0.36 and ≈ -0.45), reflecting severe catastrophic forgetting without a projection or regularization mechanism. *Split-MNIST MH.* Most methods cluster near the ceiling ($\geq$ 95%), reflecting the low difficulty of the binary per-task splits. `OGD`, `iFOPNG`, `SGD`, and `FOPNG` lead (96.1–95.6%); `ONG` and `FNG` are the notable exceptions (87% and 86%) with strongly negative BWT, indicating convergence instability on this stream.

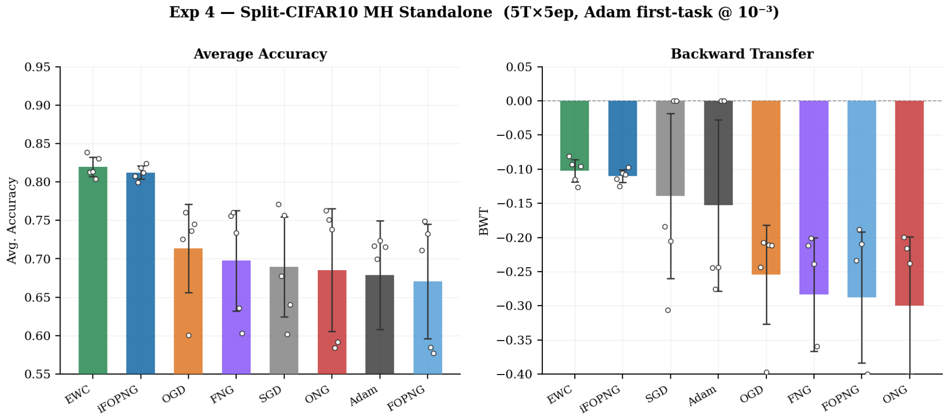

Figure 8: **Standalone target-network results — Split-CIFAR10 multi-head (5 tasks, 5 epochs per task, Adam first-task at 10^{-3}).** Left: average accuracy. Right: backward transfer. `EWC` leads at 81.9% and `iFOPNG` is a close second at 81.1%; both exhibit moderate forgetting (BWT ≈ -0.09). The remaining methods fall to 67–71% with substantially higher seed variance and stronger forgetting. `FOPNG` is the weakest projection method at 67.4%, a 13.7 pp gap below `iFOPNG` that motivates the Sub-RQ3 comparison. `Adam` (68.1%) performs comparably to the weaker projection methods, confirming that the unconstrained baseline is not dominant on harder standalone benchmarks.

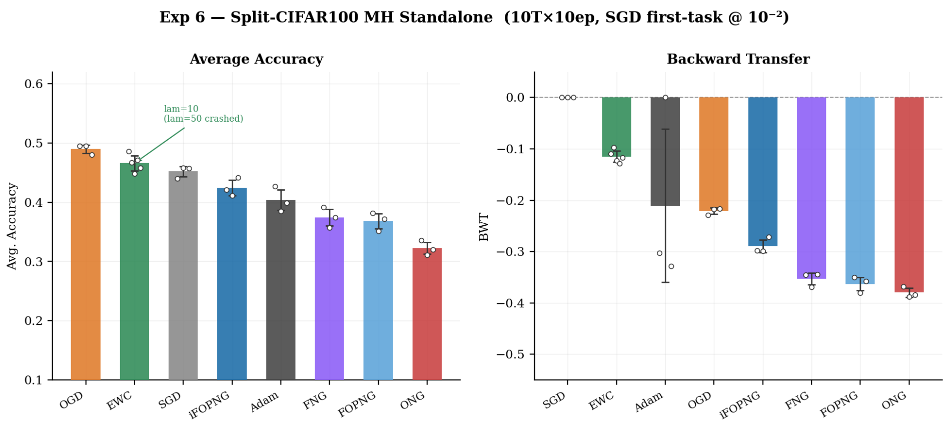

Figure 9: **Standalone target-network results — Split-CIFAR100 multi-head (10 tasks, 10 epochs per task, SGD first-task at 10^{-2}).** Left: average accuracy. Right: backward transfer. EWC was run with $\lambda = 10$; the $\lambda = 50$ configuration produced collapsed runs and is excluded. `OGD` leads at 49.2%, followed by `EWC` (λ=10, 46.8%) and `SGD` (45.4%). `iFOPNG` reaches 42.7%, a 5.6 pp advantage over `FOPNG` (37.1%), consistent with the Sub-RQ3 finding that parameter inertia provides increasing benefit as task count and Fisher overlap grow. `ONG` has the lowest average accuracy (32.3%) and the most negative BWT (≈ -0.41), indicating near-complete forgetting across the ten-task sequence. Random-chance performance is 10% (10-way classification per task).

APPENDIX C: ADDITIONAL EXPERIMENT FIGURES

This appendix collects four figures that are referenced in the Results section but omitted from the main text for space. Figure 10 provides the per-task accuracy matrix for the Split-CIFAR10 hypernetwork experiment. Figure 11 shows the full method comparison on the ten-task Split-CIFAR100 hypernetwork benchmark. Figure 12 documents the SGD first-task convergence diagnostic. Figure 13 provides the per-seed paired comparison underlying the Sub-RQ4 statistical test.

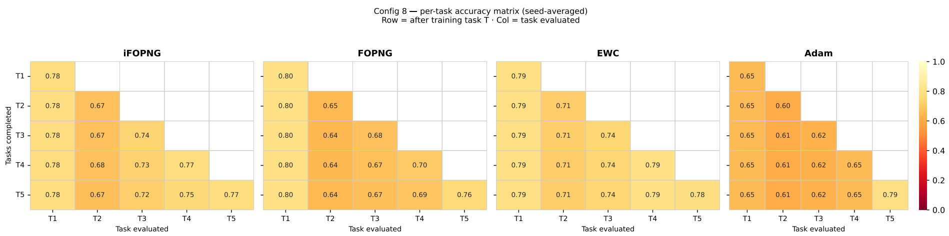

Figure 10: **Per-task accuracy matrix — Split-CIFAR10 hypernetwork, seed-averaged (Sub-RQ1).** Each matrix shows the accuracy on task j (column) after completing training through task i (row). Only the lower triangle is populated; the diagonal gives the just-trained task's accuracy and off-diagonal entries measure retention of earlier tasks. Four methods are shown: `iFOPNG`, `FOPNG`, `EWC`, and `Adam`. The T1 column is the primary retention indicator. `iFOPNG`, `FOPNG`, and `EWC` hold T1 accuracy stable at 0.78, 0.80, and 0.79 respectively across all five task transitions. `Adam` retains T1 at only 0.65 — 0.13 pp lower — indicating that the unconstrained adaptive optimizer forgets more of the first task than any of the projection or regularization methods. However, `Adam` achieves the highest forward transfer on the final task: the T5 diagonal entry is 0.79, compared to 0.77 (`iFOPNG`), 0.76 (`FOPNG`), and 0.78 (`EWC`). The average accuracy gap between `Adam` and projection methods in the main text therefore reflects weaker forward transfer to later tasks, not stronger forgetting of earlier ones — a distinction visible directly in this matrix.

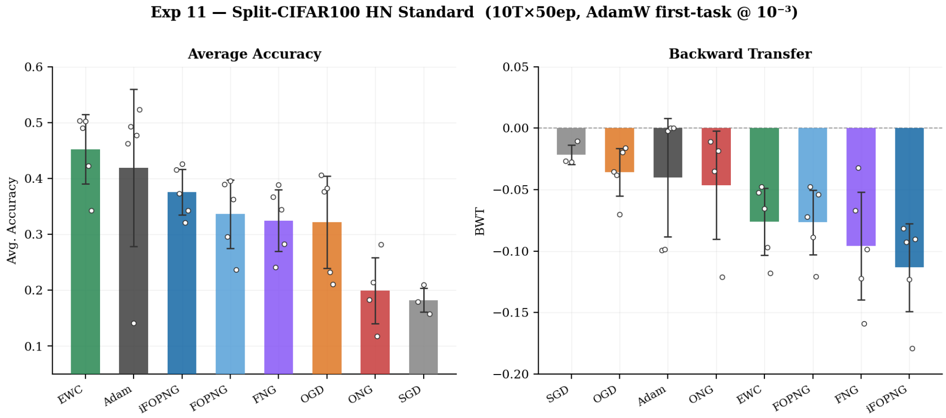

Figure 11: **Split-CIFAR100 hypernetwork — 10 tasks, 50 epochs per task, `AdamW` first-task initialisation at** 10^{-3}**.** Left: average accuracy. Right: backward transfer. Random-chance performance is 10% (10-way classification per task). `EWC` leads at 45.5%, with `Adam` second at 42.0% and `iFOPNG` third at 37.8% — a rank ordering that inverts the five-task Split-CIFAR10 pattern (where projection methods outperform `EWC` and closely trail `Adam`). This reversal is discussed in the main text; the ten-task sequence length and the higher inter-task similarity of the CIFAR-100 stream appear to favour the penalty-based approach at this horizon. Seed-level variance is elevated across all methods; notably, `Adam` contains one outlier seed at $\approx$ 14%, pulling the mean below the median. `SGD` (18.7%) and `ONG` (20.1%) barely exceed random chance, indicating near-complete forgetting on this benchmark.

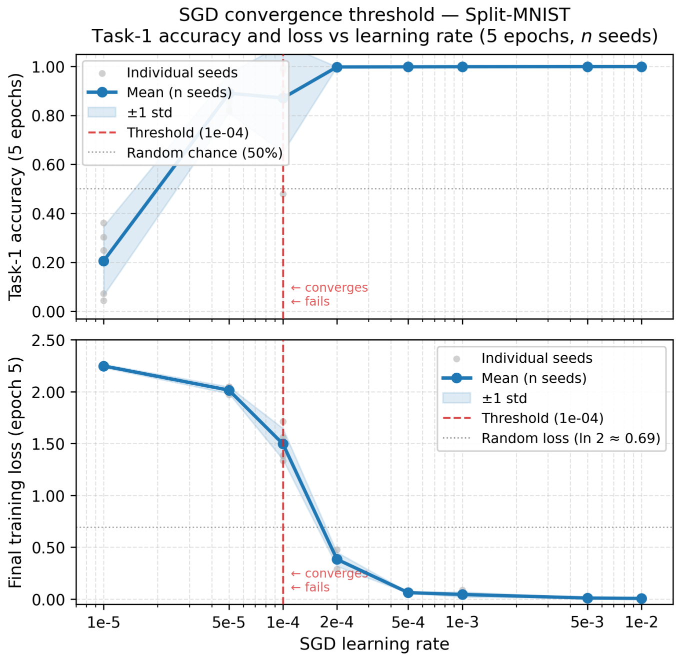

Figure 12: **SGD first-task convergence threshold — Split-MNIST (5 epochs, task-1 only).** Top: task-1 accuracy after five epochs of SGD training as a function of learning rate. Bottom: corresponding final training loss. The red dashed line at $\eta = 10^{-4}$ marks the empirical convergence threshold: below it, accuracy is at or near random chance (50%) and loss remains above 1.5; above it, the network converges reliably to near-perfect task-1 accuracy. The FOPNG paper specifies $\eta = 10^{-5}$ for first-task SGD training, which this sweep shows yields only $\approx 21\%$ accuracy — barely above random chance. The convergence threshold lies between 5×10^{-5} and 10^{-4}. All standalone experiments in this thesis use $\eta = 10^{-3}$, which achieves near-perfect first-task convergence and provides a clean initial projection subspace.

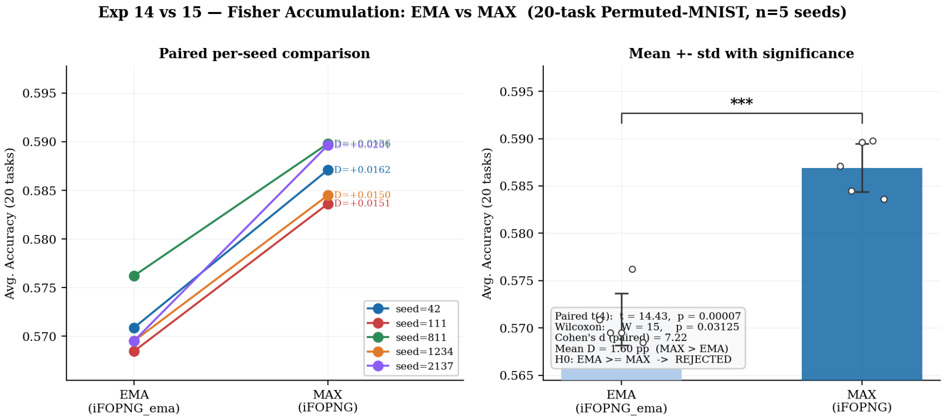

Figure 13: **EMA vs. MAX Fisher accumulation — per-seed statistical comparison, 20-task Permuted-MNIST, $n = 5$ seeds (Sub-RQ4).** Left: paired slope plot. Each line connects the final average accuracy of one seed under EMA (left) and MAX (right); the annotated D values give the per-seed difference. Right: mean $\pm$ std bar chart with the paired t-test significance bracket. Every seed shows a positive slope (MAX $>$ EMA), confirming that the direction of the effect is consistent without exception. The per-seed differences range from $+1.50$ to $+1.65$ pp. Formal tests: paired $t(4) = 14.43$, $p < 0.001$; Wilcoxon signed-rank $W = 15$, $p = 0.031$; Cohen's $d = 7.22$. The null hypothesis that EMA performs at least as well as MAX is rejected by both tests. As discussed in the main text, the effect is statistically unambiguous but practically negligible (1.60 pp mean difference on a benchmark where both strategies lose more than 30 pp over the full sequence).

APPENDIX D: FULL BENCHMARK METHOD COMPARISON

This appendix provides a complete accuracy reference for all eight methods across every benchmark evaluated in this thesis. Figures 14 and 15 present side-by-side grouped bar charts contrasting standalone and hypernetwork performance on Split-CIFAR10 and Split-MNIST respectively, allowing direct reading of each method's accuracy and BWT under both settings. Table 1 collects mean average accuracy across all seven benchmarks in a single table for quick cross-benchmark reference. Following standard practice in the continual learning literature, results are reported as mean $\pm$ standard deviation over seeds without formal pairwise significance tests; formal statistical comparison is reserved for the targeted iFOPNG vs. FOPNG analysis in Sub-RQ3 (Section 5.3) and the EMA vs. MAX analysis in Sub-RQ4 (Section 5.4).

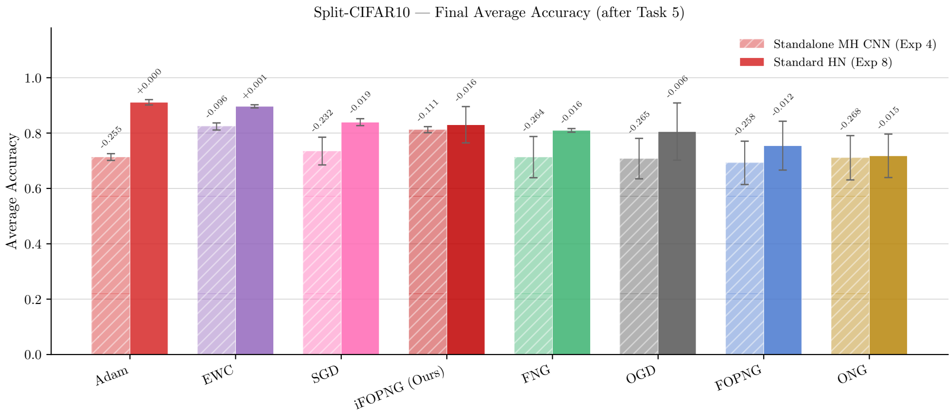

Figure 14: **Split-CIFAR10 — all methods, standalone MH CNN (left) vs. standard hypernetwork (right).** Bars show mean final average accuracy after all five tasks; BWT is annotated on each bar. The left panel corresponds to the standalone benchmark in Appendix B (Figure 8); the right panel corresponds to the main-text hypernetwork experiment (Figure 1). On the standalone benchmark `EWC` and `iFOPNG` lead (81.9% and 81.1%), while `FOPNG` is the weakest projection method (67.4%). In the hypernetwork setting the ranking partially reorders: `Adam` extends its lead (90.1%) and `iFOPNG` remains the best projection method (85.6%), but `ONG` drops substantially relative to its standalone position. The BWT annotations confirm that the hypernetwork's functional regularizer suppresses forgetting across all methods relative to their standalone counterparts.

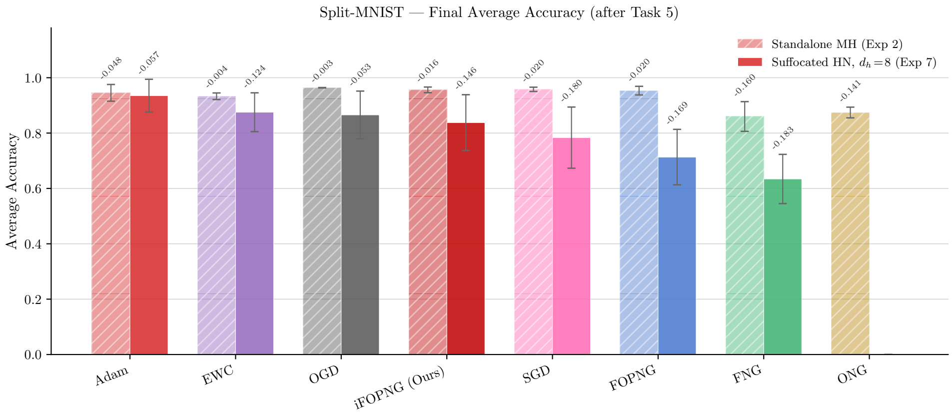

Figure 15: **Split-MNIST — all methods, standalone MH CNN (left) vs. suffocated hypernetwork at $d_h = 8$ (right).** Bars show mean final average accuracy after all five tasks; BWT is annotated on each bar. The left panel corresponds to Appendix B (Figure 7, bottom row); the right panel corresponds to the main-text suffocated-regime experiment (Figure 2). On the standalone benchmark most methods cluster near 96%, with `ONG` and `FNG` as notable exceptions. Under extreme compression ($d_h = 8$) the ranking restructures substantially: `Adam` maintains high accuracy (93.4%) while Fisher-based projection methods (`iFOPNG`, `FOPNG`, `FNG`) collapse to 55–70% despite near-zero BWT, revealing the plasticity failure described in the main text. `OGD` is the most resilient projection method (86.6%), consistent with its Fisher-free projection mechanism. Note that several methods completed fewer than three seeds in the suffocated regime; bars with low seed counts should be interpreted with caution.

Table 1: **Mean average accuracy (%) across all benchmarks and methods.** Values are seed-averaged final average accuracy after all tasks. Standalone benchmarks use multi-head target networks; HN benchmarks use the chunked hypernetwork architecture. †MNIST HN is the suffocated regime ($d_h = 8$). ‡CIFAR-100 MH EWC used $\lambda = 10$; $\lambda = 50$ collapsed. **Bold**: best per column. ***Bold italic***: `iFOPNG`. —: insufficient seeds or unavailable.

	Standalone				*Hypernetwork*		
Method	Perm-MNIST	MNIST MH	CIFAR-10 MH	CIFAR-100 MH‡	MNIST HN (d_h=8)†	CIFAR-10 HN	CIFAR-100 HN
iFOPNG (ours)	***86.1***	95.9	81.2	42.7	69.9	85.6	37.8
FOPNG	85.8	95.3	67.4	37.1	59.9	77.5	33.8
EWC	83.8	93.2	**81.9**	46.8	84.5	81.5	**45.5**
OGD	80.8	**96.1**	71.4	**49.2**	**86.6**	78.3	32.3
FNG	69.2	86.1	70.0	37.5	54.8	80.9	32.5
ONG	63.0	87.1	68.8	32.3	—	67.2	20.1
SGD	84.9	**96.1**	68.9	45.4	79.8	84.1	18.7
Adam	68.2	95.1	68.1	40.5	**93.4**	**90.1**	42.0
Random chance	50.0	50.0	50.0	10.0	50.0	50.0	10.0

APPENDIX E: HYPERPARAMETER TABLES

All values are sourced directly from the W&B run configs (CSV logs) unless marked ? (inferred from thesis text or figure annotations). Entries marked † are flagged in Section 8. `max_ep` is the early-stopping patience; `ep/task` is epochs per task; `max_dirs` is the gradient memory column budget. Values that are constant across all experiments in a group are stated in the relevant caption rather than repeated in every row.

E.1 Standalone Target-Network Experiments

Table 2: **Standalone training hyperparameters.** All rows: `model = TargetNetwork`, `normalize = False`, `regulizer = False`, `damping = 0.01`, $n = 5$ seeds.

Benchmark	Tasks	ep/task	max_ep	First opt	First LR	LR	Batch	Fisher samp.
Permuted-MNIST	5	10	5	SGD	10^{-3}	10^{-4}	10	60 000
Split-MNIST MH	5	10	5	SGD	10^{-5}	10^{-5}	10	12 000
Split-CIFAR10 MH	5	10	5	Adam	10^{-3}	10^{-3}	10	1 024
Split-CIFAR100 MH	10	10	10	SGD	10^{-2}	5×10^{-3}	10	1 024

Table 3: **EWC λ for standalone experiments.** Projection methods and Adam/SGD use $\lambda = 0.001$ (minimal). ‡CIFAR-100 MH: $\lambda = 50$ collapsed and is excluded; $\lambda = 10$ is used (annotated in Figure 9). ?Other EWC values inferred from the set of distinct `lam` values in each experiment's logs.

Benchmark	EWC λ	Log values present
Permuted-MNIST	$10^{?}$	$\{10,\ 0.01,\ 0.001\}$
Split-MNIST MH	$0.0005^{?}$	$\{400,\ 0.001,\ 0.0005\}$
Split-CIFAR10 MH	10 or $50^{?}$	$\{100\,000,\ 50,\ 10,\ 0.001\}$
Split-CIFAR100 MH	$10^{‡}$	Confirmed; $\lambda = 50$ excluded

E.2 Hypernetwork Experiments

All HN experiments: `model = HyperNetwork`, `regulizer = True`, $\beta = 0.001$ (`lam = 0.001`), $\eta = 10^{-3}$, $n = 5$ seeds.

Table 4: **Hypernetwork architecture and training parameters.** `c_emb` = chunk embedding dim; `t_emb` = task embedding dim. †Exp 407: `first_task_opt = None` in logs; likely AdamW by code default (see E.6). †Exp 411: extracted from the 35 completed runs at chunk = 6 000, $d_h = 32$, `normalize=True`.

Role / Benchmark	Tasks	ep/task	max_ep	First opt	d_h	chunk	c_emb	t_emb	grads / task
CIFAR-10 HN (main)	5	10	50	AdamW	32	6 000	16	16	80
CIFAR-100 HN†	10	10	50	AdamW	32	6 000	16	16	80
MNIST HN ($d_h = 8$, suffocated)†	5	10	15	AdamW	8	64	32	32	100
dh sweep $d_h = 4$	5	10	15	AdamW	4	64	32	32	80
dh sweep $d_h = 8$	5	10	15	AdamW	8	64	32	32	100
dh sweep $d_h = 16$	5	10	15	AdamW	16	64	32	32	80

E.3 Normalization Ablation

All three conditions share the same architecture: Split-CIFAR10, 5 tasks, `max_ep = 50`, chunk = 6 000, $d_h = 32$, batch = 32, AdamW first-task at $\eta = 10^{-3}$, `iFOPNG` only.

Table 5: **Normalization ablation conditions (Sub-RQ2).** The CSVs for conditions 1 and 2 also contain pilot runs at chunk = 256, $d_h = 64$; these are excluded. Only chunk = 6 000, $d_h = 32$ runs are used, confirmed by the reported accuracy values.

Condition	`normalize`	QR on G	Max-scale $\hat{F}$	Seeds	Accuracy	BWT
No normalization	`False`	×	×	3	72.2%	−0.070
Gradient-only	`True`	✓	×	3	68.9%	−0.060
Full normalization	`True`	✓	✓	3	85.8%	−0.013

E.4 Sub-RQ4: Fisher Accumulation Operators

Both conditions: 20-task Permuted-MNIST, `TargetNetwork`, `normalize = False`, batch = 10, $n = 5$ seeds.

Table 6: **Sub-RQ4 accumulation parameters. Note:** the Discussion incorrectly states $\alpha = 0.3$; the confirmed value from the run logs is $\alpha = 0.5$.

Operator	Tasks	ep/task	max_ep	LR	Fisher samp.
MAX (`iFOPNG`)	20	10	5	10^{-4}	60 000
EMA (`iFOPNG_ema`, $\alpha = 0.5$)	20	10	5	10^{-4}	60 000

E.5 SGD Convergence Threshold Diagnostic

The SGD threshold diagnostic (Figure 12) sweeps $\eta \in \{10^{-5}, 5\times10^{-5}, 10^{-4}, 2\times 10^{-4}, 5\times10^{-4}, 10^{-3}, 5\times10^{-3}, 10^{-2}\}$ on Split-MNIST SH (`TargetNetwork`, 1 task, 5 epochs). Convergence threshold: $\eta = 10^{-4}$. All standalone experiments in this thesis use $\eta = 10^{-3}$.

E.6 Notes on Confounds and Uncertainties

EXP 407 `first_task_opt = none`. The run config records `first_task_opt = None` and `first_task_lr = None`. The figure title annotates "AdamW first-task," likely reflecting a code-level default: Exps 412 and 413 (same architecture, added later) explicitly log `adamw`, suggesting Exp 407 predates that config field.

EXP 411 PARAMETER HETEROGENEITY. The CIFAR-100 HN CSV contains 222 runs across a wide range of `chunk_size`, `hyper_hidden_dim`, and normalization settings — an exploratory search prior to settling on the published architecture. The 35 completed runs at chunk = 6 000, $d_h = 32$, `normalize=True` are used for the reported results; all other runs are pilots.

NORMALIZATION ABLATION PILOT RUNS. Exps 409 and 410 each contain additional runs at chunk = 256, $d_h = 64$ (an earlier pilot) alongside the published chunk = 6 000, $d_h = 32$ runs. The pilot runs reach near-random accuracy (≈50–61%) and are excluded; the ablation is architecturally controlled.

EWC λ IDENTIFICATION. The `lam` field is shared between EWC's λ and the projection damping term, so per-method EWC values cannot be extracted from the CSV without row-level method filtering; Table 3 values are inferred from the set of distinct `lam` values per experiment.